

Additional Practice

2.05

- 1.** A mother buys lunch boxes for each of her 4 children. She pays a total of \$52. If n represents the number of lunchboxes and c represents the total cost, in dollars, which equation gives the relationship between n and c ?

- A. $n = 13c$
- B. $c = n + 13$
- C. $c = 13n$
- D. $n = c + 13$

Problems 2–4: A line cook at a restaurant uses 8 tomatoes to make 12 bowls of salad. Assume all tomatoes and salads are about the same size.

- 2.** How many bowls of salads can be made with 1 tomato? Show or explain your thinking.
- 3.** Write an equation that represents the relationship between the number of tomatoes used, x , and the number of bowls of salads that can be made, y .
- 4.** If the line cook has 24 tomatoes, how many salad bowls can she make?

Problems 5–6: A hardware store sells copper wire by the foot. The equation $c = 1.27f$ represents the cost c , in dollars, of a copper wire that is f ft long.

- 5.** What does the 1.27 represent in this situation?
- 6.** What is the cost of 50 feet of copper wire? Show or explain your thinking.

Problems 7–8: On its way from Orlando, Florida to Los Angeles, California, a plane flew at a constant speed over Baton Rouge, Houston, San Antonio, and Phoenix.

7. This table shows the flight time and distance traveled for each segment on the flight. Complete the table.

Segment	Time (hr)	Distance (mi)	Speed (mph)
Baton Rouge to Houston	0.6	268	
Houston to San Antonio	2.3		
San Antonio to Phoenix		980	

8. Let t represent the time in hours and d represent the distance in miles. Write an equation that represents the distance traveled for t hours.

9. Rodney runs at a constant speed at a recent cross country event. He ran $3\frac{1}{10}$ miles in $\frac{3}{5}$ of an hour. How far does Rodney run in one hour at this same speed?

Additional Practice

ACC6.5.06

Problems 1–2: The relationship between a distance in feet, y , and the same distance in miles, x , is represented by the equation $y = 5280x$.

- Complete the table.
- Is there a proportional relationship between a distance in feet and the same distance in miles? Circle one.

Yes No

Explain your thinking.

Distance (mi), x	Distance (ft), y
1	
4	
	15,840
	52,800

Problems 3–6: Determine whether or not each relationship is proportional. Circle one.

- | | | |
|--|--------------|------------------|
| 3. The remaining slices of bread, s , from a loaf of 22 slices after x number of slices have been eaten: $22 - x = s$ | Proportional | Not proportional |
| 4. The number of pencils, x , each student gets after p pencils are shared equally among 5 students: $x = \frac{p}{5}$ | Proportional | Not proportional |
| 5. The total cost, c , after 20% tip is added to the restaurant bill, b : $1.20b = c$ | Proportional | Not proportional |
| 6. The volume, V , of a rectangular prism, whose height is 8 inches and whose base is a square with side lengths of s inches: $V = 8s^2$ | Proportional | Not proportional |

Problems 7–10: Determine whether each relationship is proportional. Explain your thinking.

7.

x	y
3	4.5
4	6
6	9

8. $y = 1.7x - 2$

9. The weight of a case of standard 20-ounce bottles of water and the number of bottles of water

10. The weight of a case of different-sized bottles of water (where each bottle weighs a different amount) and the number of bottles of water

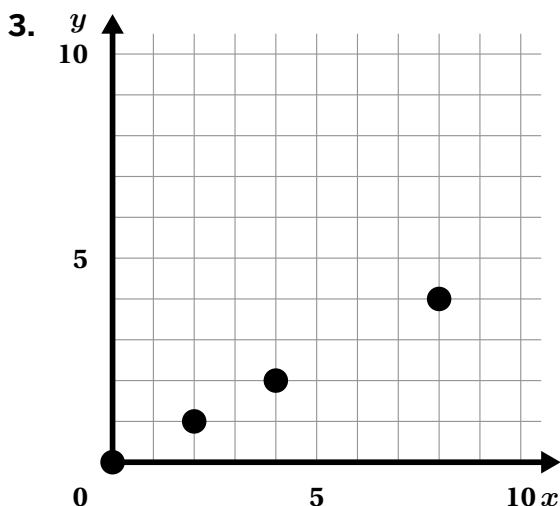
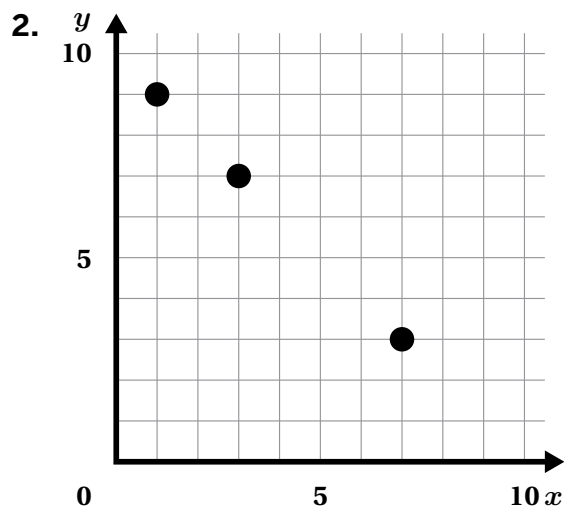
Additional Practice

ACC6.5.07

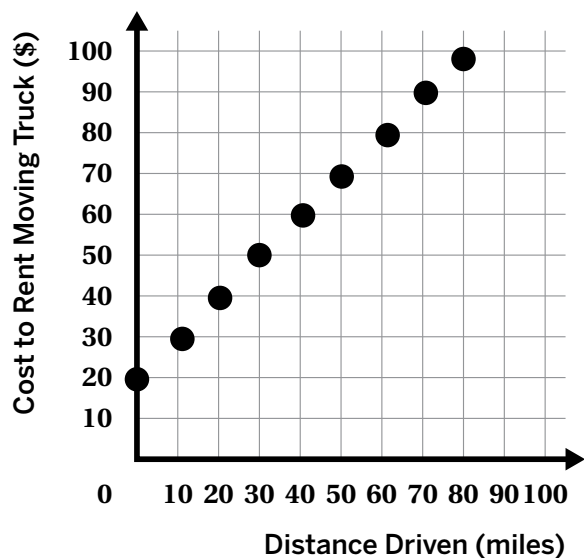
1. Which of the following statements about the graph of a proportional relationship are true? Select *all* that apply.

- ☐ A. The points on the graph form a curve.
- ☐ B. The points on the graph form a straight line.
- ☐ C. The graph goes through the origin.
- ☐ D. The graph must be a solid line.
- ☐ E. The graph could be a series of points.

Problems 2–3: Determine if each graph represents a proportional relationship. Explain your thinking.



4. The graph shows the relationship between the number of miles a moving truck has driven and the total cost of renting the truck. Explain why this relationship is not proportional based on the graph.



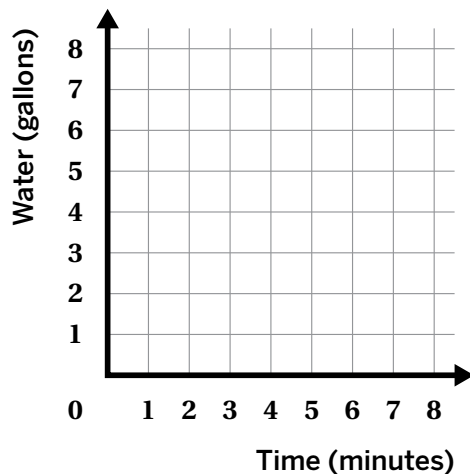
5. Complete the table so that it shows a proportional relationship.

<i>x</i>	0	2	4	6
<i>y</i>			12	

6. Fiona earned \$50.00 for mowing 4 lawns. Given the pay rate stays the same, how much will Fiona earn if she mows 6 lawns? Explain your thinking.
7. A bathtub is being filled at a rate of $1\frac{1}{3}$ gallons of water per minute. The table shows how much water is in the bathtub after several minutes have passed.

Time (minutes)	1	2	3	4
Water (gallons)	$1\frac{1}{3}$	$2\frac{2}{3}$	4	$5\frac{1}{3}$

Graph the ordered pairs to determine whether the relationship between time and the amount of water in the bathtub is proportional. Explain your thinking.



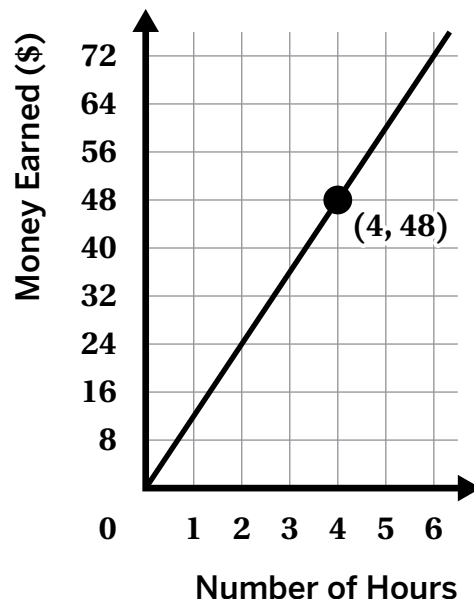
Additional Practice

ACC6.5.08

1. Here is a graph that shows a proportional relationship between the hours Andre babysits and the total amount of money he earns.

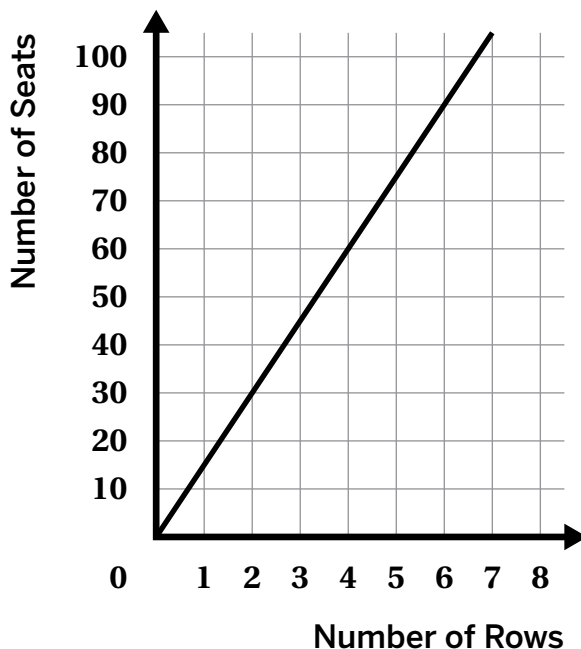
After 4 hours, Andre earns \$48.

- What is a constant of proportionality in this relationship?
- Write an equation that represents this relationship, where x represents the number of hours and y represents the money earned.



2. The graph shows the number of rows and the number of seats in a school auditorium.

- Determine a constant of proportionality for this relationship, then explain its meaning.
- Label one place you see that constant of proportionality on the graph.

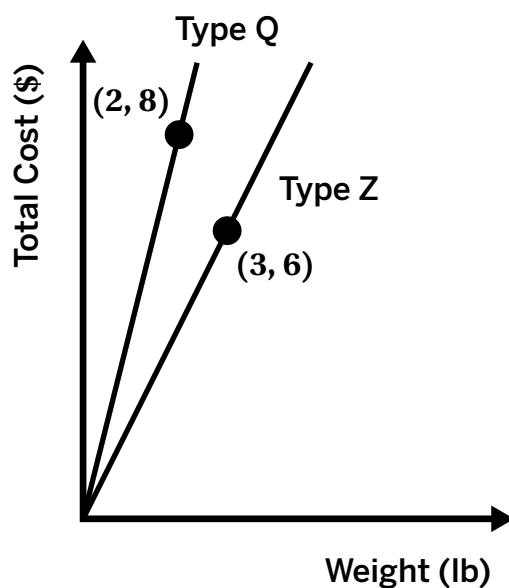


3. The graph shows the cost for two different types of bananas. Which type of banana has a higher cost per pound? Circle one.

Type Q

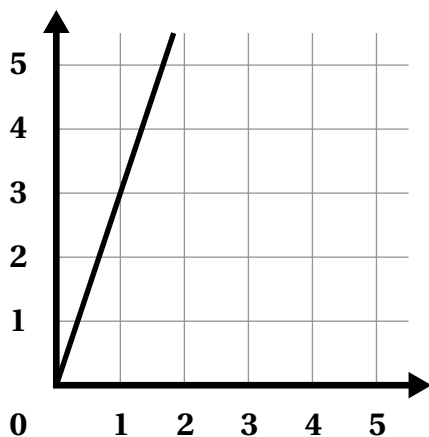
Type Z

Explain your thinking.

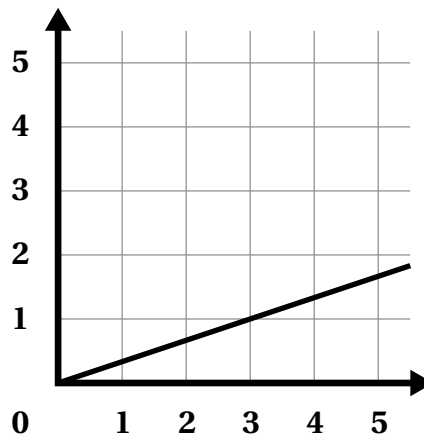


4. Which of these relationships has the same constant of proportionality as the equation $y = \frac{1}{3}x$? Select *all* that apply.

☐ A.



☐ B.



☐ C.

x	y
0	0
1	3
2	6
3	9

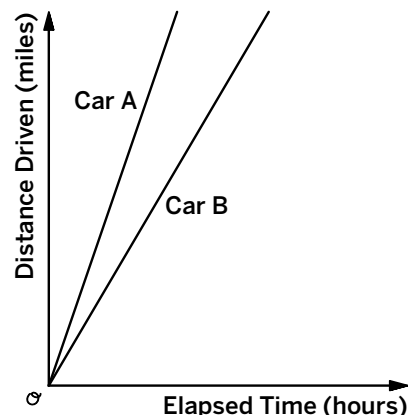
☐ D.

x	y
0	0
3	1
6	2
9	3

Additional Practice

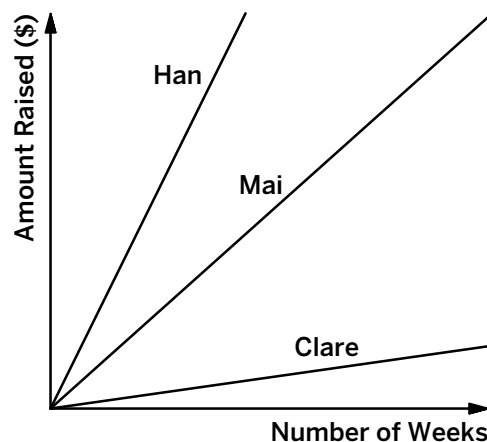
ACC6.5.09

1. Two cars are traveling. The graph shows the distance driven over time. Does Car A travel faster or slower than Car B? Explain your thinking.



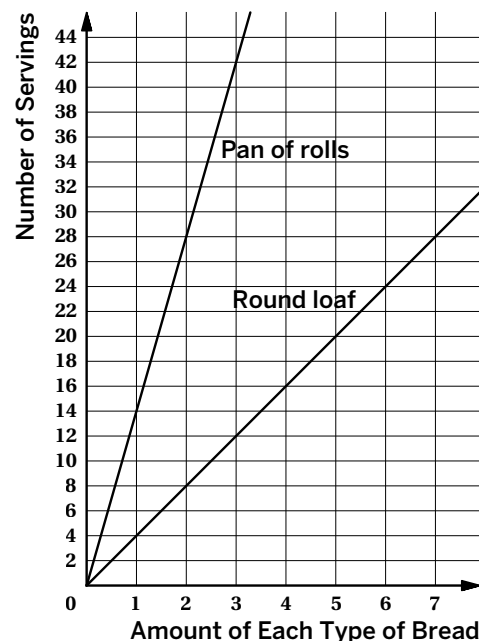
2. The graph shows the amount of money raised by Han, Mai, and Clare at a school fundraiser over several weeks.

List the names in order of who raised money at the slowest rate to the fastest rate.



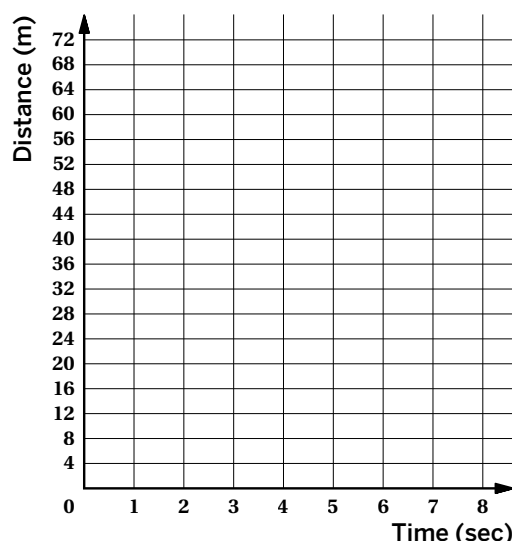
3. A bakery sells round loaves of bread and pans of rolls. The graph shows the relationship between the amount of each type of bread and the number of servings. Select *all* the statements that are true based on the graph.

- ☐ A. Both lines show a proportional relationship between the amount of each type of bread and the number of servings.
- ☐ B. 2 pans of rolls can serve 28 people.
- ☐ C. There are 4 servings in 16 round loaves.
- ☐ D. There are more servings in a round loaf than in a pan of rolls.
- ☐ E. A round loaf serves less people than a pan of rolls.
- ☐ F. The line that represents the number of servings in a pan of rolls has a greater constant of proportionality.

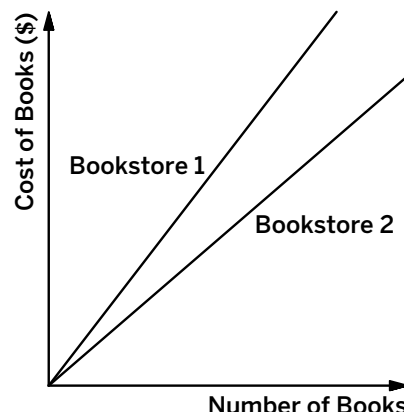


4. Bard and Priya race each other on the school track. Priya takes 5 seconds to run 35 meters and Bard takes 4 seconds to run 32 meters. Both students run at a constant rate.

- Graph the two lines that represent the distance that Bard and Priya run on the track. Label each line.
- Write an equation that represents Bard's and Priya's races. Use t for time and d for distance.
- Which student won the race?

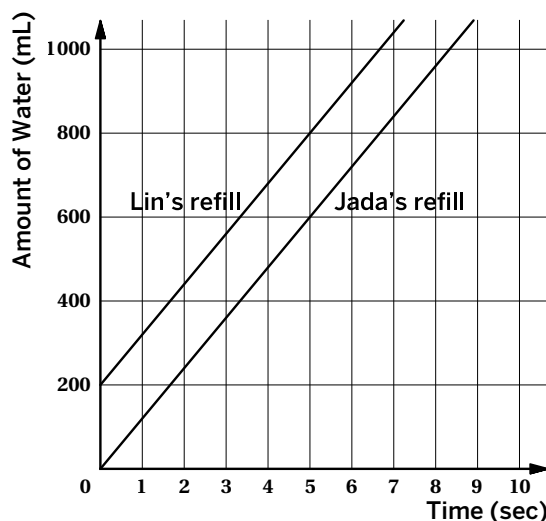


5. A teacher needs to order a class set of books for his class. The graph shows two lines that represent the total cost of the books at two different bookstores. From which bookstore should the teacher purchase his class set of books? Explain your thinking.



6. Jada and Lin refill their water bottles at a drinking fountain after basketball practice. The graph represents the amount of water in their bottles as they fill them up at the drinking fountain.

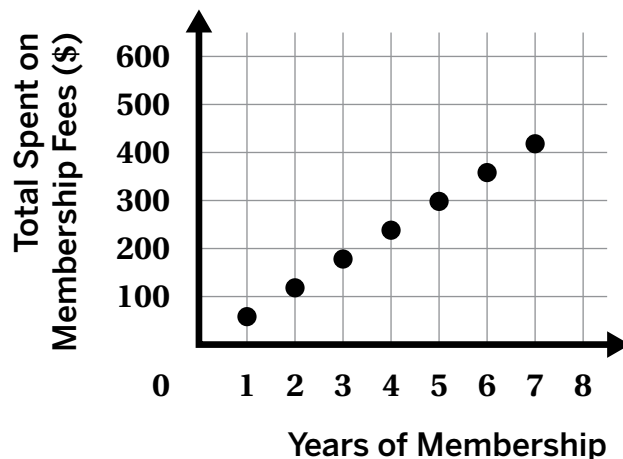
- Which line represents a proportional relationship? Explain your thinking.
- What does the point $(0, 0)$ mean on the line representing Jada's refill?
- What does the point $(0, 200)$ mean on the line representing Lin's refill?
- Which line represents the bottle that was filled at a faster rate? Explain your thinking.



Additional Practice

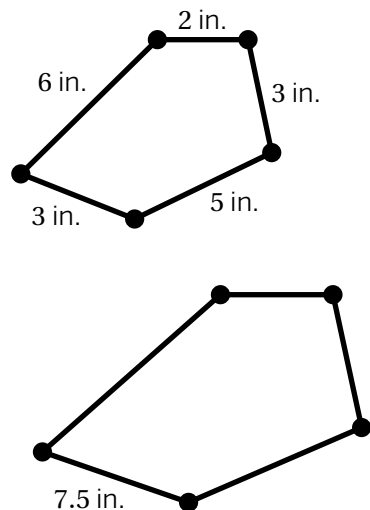
ACC6.5.12

Problems 1–4: A wholesale store charges an annual membership fee for its customers to shop there. The graph shows the relationship between the total amount spent on membership fees and the number of years that a customer is a member.



1. Explain how the graph shows that the relationship between the number of years a customer has been a member and the total amount of money spent on membership fees is proportional.
2. Calculate a constant of proportionality for this relationship.
3. Write an equation that relates years of membership, m , to the total amount of money spent on membership fees, C .
4. If a customer paid a total amount of \$840 in membership fees, for how many years have they been a customer?

5. These polygons are scaled copies. Determine the perimeter of the larger polygon. Show or explain your thinking.



Problems 6–9: The relationship between the number of guests g invited to a wedding reception and the number of tables t that are needed to accommodate them is proportional.

6. Complete the table.

Number of guests, g	Number of tables, t
80	10
	15
200	
	35

7. What are the two constants of proportionality for this relationship?
8. Write two equations using the constants of proportionality from Problem 7.
9. Use one of your equations from Problem 8 to determine how many tables will be needed to accommodate 300 guests.

Additional Practice

ACC6.5.13

- The tires on a sports car are 2.5 feet tall. Which part of a circle does this measurement represent?
 - Diameter
 - Radius
 - Circumference
 - Area
- Determine the circumference of a circle with a diameter of 8 inches.

- Elena measured the diameter and circumference of several circular objects and recorded her measurements in the table. Fill in the missing values of the table. Round your answers to the nearest whole number.

	Radius (mm)	Diameter (mm)	Circumference (mm)
Button	6		
Quarter			75
Cucumber Slice		38	
Ring			52

- Elena is painting on a circular canvas. The circumference of the canvas is 16π inches. Determine the diameter of the canvas Elena is painting on.

Problems 5–7: Determine whether each pair of measurements could be a reasonable approximation for the diameter and circumference of a circle. Show or explain your thinking.

5. 11 inches and 45 inches

6. 20 inches and 47 inches

7. 16 inches and 50 inches

8. Complete the table with a possible length for each diameter or circumference, given the other measurement. Verify your answers by calculating the ratio of circumference to diameter for each pair of lengths.

Diameter	Circumference	$\frac{\text{Circumference}}{\text{Diameter}}$
9		
48		
	116	

Additional Practice

ACC6.5.14

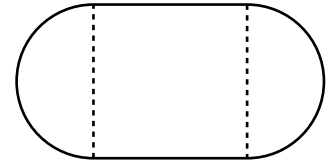
- Determine the *exact* measurements of the diameter and the radius of a circle if the circumference is 7π .
 - Radius: 7, Diameter: 14
 - Radius: 7, Diameter: 3.5
 - Radius: 3.5, Diameter: 7
 - Radius: 1.75, Diameter: 3.5
- Determine the exact measurements of the diameter and the radius of a circle if the circumference is 24.
 - Radius: 12, Diameter: 24
 - Radius: $\frac{6}{\pi}$, Diameter: $\frac{12}{\pi}$
 - Radius: $\frac{24}{\pi}$, Diameter: $\frac{48}{\pi}$
 - Radius: $\frac{12}{\pi}$, Diameter: $\frac{24}{\pi}$
- For each measurement, determine whether it represents the *radius*, *diameter*, or *circumference*. Record the measurement in the table, then determine the *exact* lengths for the other two measurements of the circle.

	Radius	Diameter	Circumference
The decorative border around a watch face measures 85 mm.			
The center to the edge of a circular pond measures 19 m.			
The length from edge to edge through the center of a vinyl record measures 25.4 cm.			

- For each measurement, determine whether it represents the *radius*, *diameter*, or *circumference*. Record the measurement in the table, then determine the *exact* lengths for the other two measurements of the circle.

	Radius	Diameter	Circumference
The center to the edge of a quarter measures 12 mm.			
The rubber around a bike tire measures 86 in.			
The length across a circular slice of cucumber measures 1.3 in.			

5. A semicircle is joined to a square with side lengths of 8 units. Noah tried to determine the perimeter of the resulting shape. What was Noah's mistake? Explain your thinking.



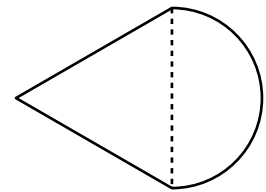
Noah's work:

2 semicircles + square

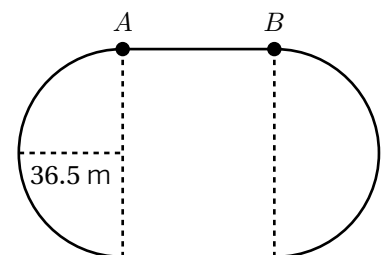
1 circle + square

$$\begin{aligned} \pi \cdot d + 4 \cdot s \\ = \pi(8) + 4(8) \\ = 8\pi + 32 \end{aligned}$$

6. A semicircle is joined to an equilateral triangle with side lengths of 14 units. Determine the exact perimeter of the resulting shape. Show or explain your thinking.



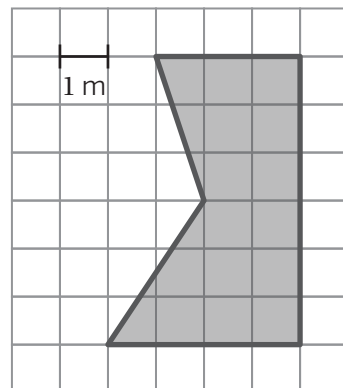
7. A track is in the shape of a rectangle with a semicircle on each end. If the distance around the entire track is 400 meters, determine the exact distance from point A to point B. Show or explain your thinking.



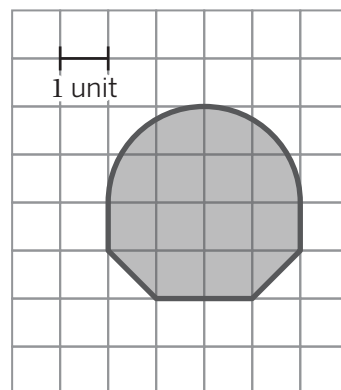
Additional Practice

3.05

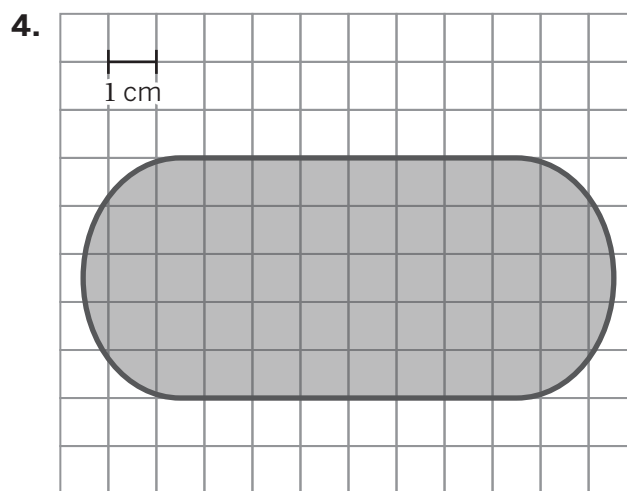
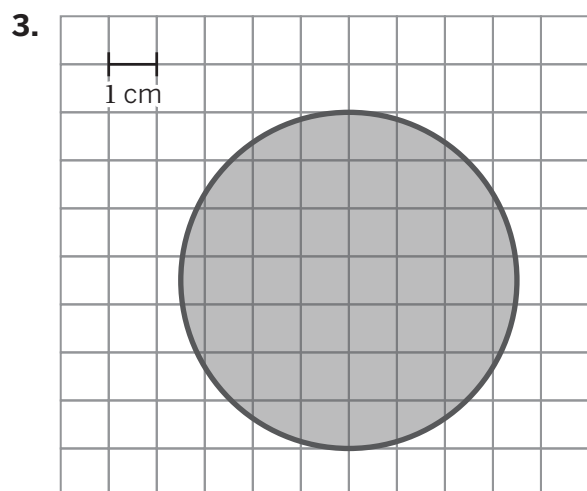
1. What is the area of this shape? Show your thinking.



2. Here is a diagram of two squares and a shape. Explain why the area of the figure is more than 8 square units but less than 16 square units.

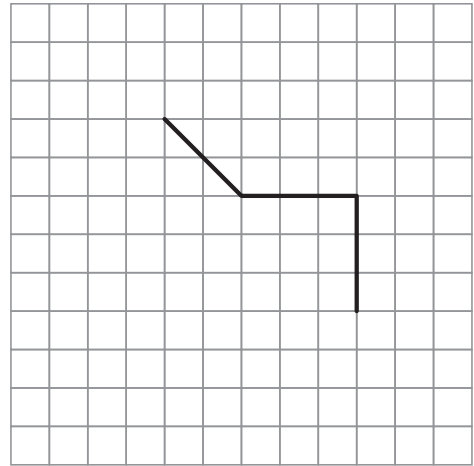


Problems 3–4: Estimate the area of each shape.

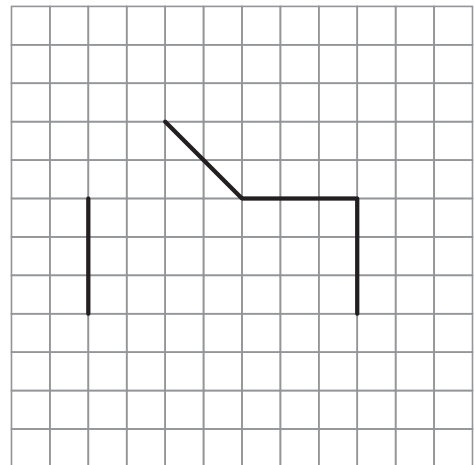


Problems 5–6: Priya started drawing a polygon.

5. Complete Priya's drawing so that the polygon has an area of 25 square units.



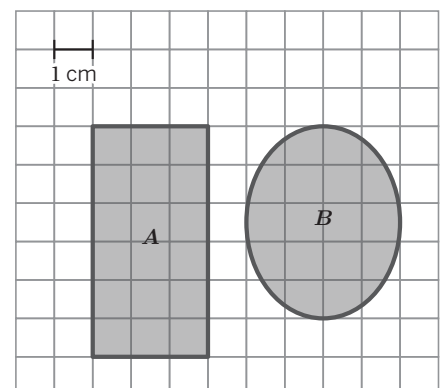
6. Complete Priya's drawing in a different way so that the polygon has an area of 25 square units.



7. Which shape has a larger area? Circle one.

Shape *A* Shape *B* They are the same.

Show or explain how you know.



Additional Practice

ACC6.5.16

1. A circle's radius is 5 cm. Which of the following is *true*?

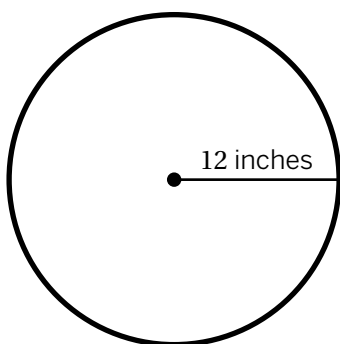
A. The circle's area is exactly $5\pi \text{ cm}^2$.	B. The circle's area is exactly $25\pi \text{ cm}^2$.
C. The circle's area is exactly $10\pi \text{ cm}^2$.	D. The circle's area is exactly $15\pi \text{ cm}^2$.

2. A circle's diameter is 12 in. Which of the following is *true*?

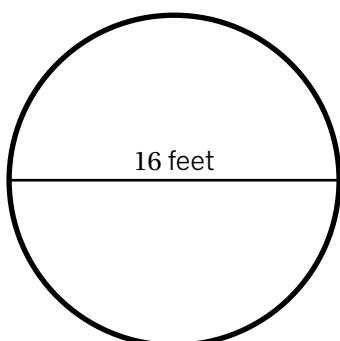
A. The circle's circumference is exactly $36\pi \text{ in}$.	B. The circle's area is exactly $144\pi \text{ in}^2$.
C. The circle's area is exactly $6\pi \text{ in}^2$.	D. The circle's area is exactly $36\pi \text{ in}^2$.

Problems 3–4: Find the exact area of each circle.

3.



4.



5. A circle has a diameter of 102 inches. What is the *exact* area of the circle, in terms of π ? Show or explain your thinking.

Problems 6–7: The radius of Mars is approximately 3,390 kilometers. The Martian dichotomy divides the northern and southern hemispheres.

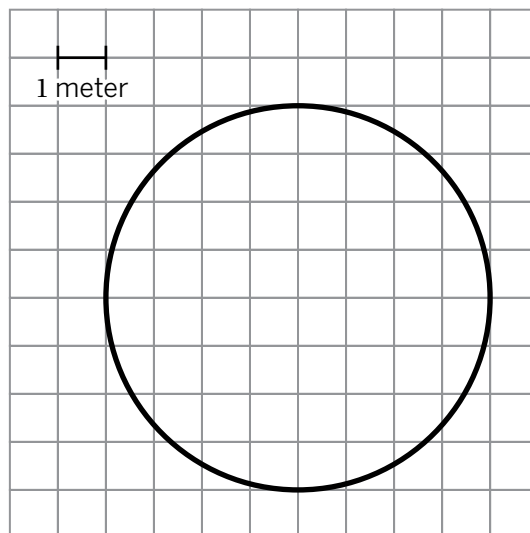
6. Is the circumference of a circle or the area of a circle more useful for finding the length of the Martian dichotomy?

7. What is the length of the Martian dichotomy?

Additional Practice

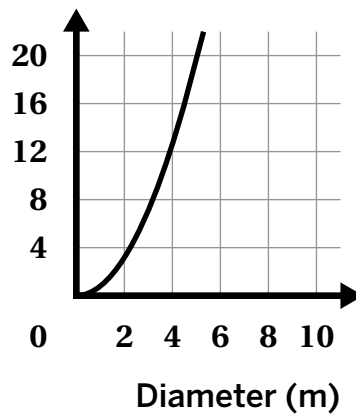
ACC6.5.17

Problems 1–2: Here is a diagram of a circle. It represents a circular road in a town. Within the circular road is a park.



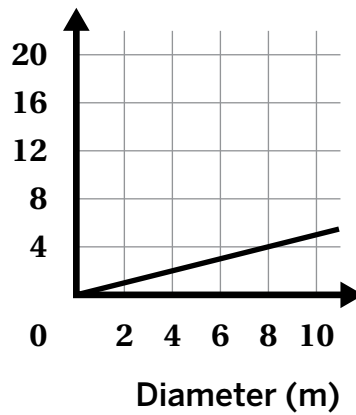
1. Estimate the area of the park. Show or explain your thinking.
2. Estimate the length of the circular road. Show or explain your thinking.
3. Are the radius and area measurements of a circle proportional to each other?
 Yes No Maybe
4. Write an equation that relates radius and diameter.

5. Circle the relationship represented by this graph.



- | | |
|------------------------|-------------------------------|
| A. Radius vs. diameter | B. Circumference vs. diameter |
| C. Area vs. diameter | D. Circumference vs. area |

6. Circle the relationship represented by this graph.



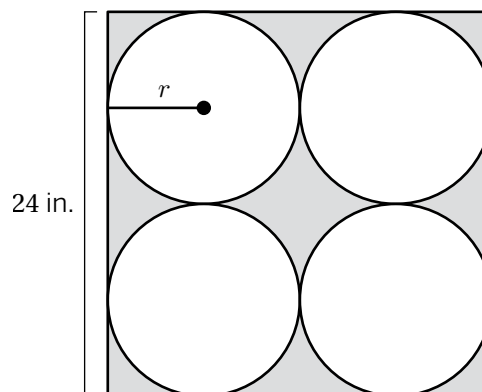
- | | |
|------------------------|-------------------------------|
| A. Radius vs. diameter | C. Circumference vs. diameter |
| B. Area vs. diameter | D. Circumference vs. area |

Additional Practice

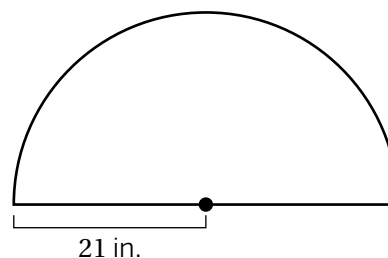
ACC6.5.18

1. Four circles are arranged in a square as shown. What is the radius of each circle?

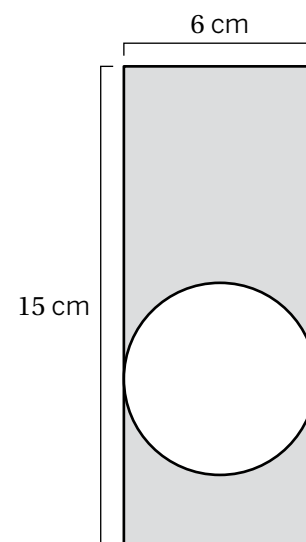
- A. $r = 3$ inches
- B. $r = 4$ inches
- C. $r = 6$ inches
- D. $r = 12$ inches



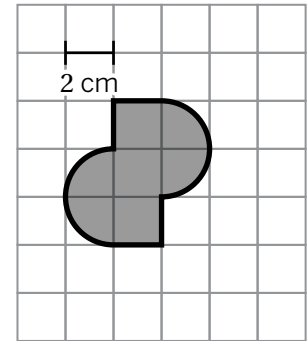
2. Calculate the area of the semicircle. Show or explain your thinking.



3. Calculate the area of the shaded region. Show or explain your thinking.



4. Determine the area of this shape.



5. A circle with a 32-inch diameter is folded in half and then folded in half again in the other direction. What is the *exact* area of the resulting shape? Write your answer in terms of π . Show or explain your thinking.
6. A circle with a 12-inch long diameter is folded in half and then folded in half again. What is the exact area of the new shape? Write your answer in terms of π . Show or explain your thinking.

Additional Practice

ACC6.6.01

1. Use benchmark percentages to help you determine each percent of 400.

a What is 50% of 400?

b What is 10% of 400?

c What is 100% of 400?

d What is 25% of 400?

2. Use benchmark percentages to help you determine each percent of 150.

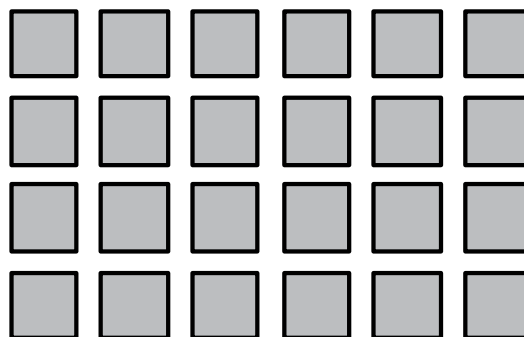
a What is 50% of 150?

b What is 10% of 150?

c What is 100% of 150?

d What is 25% of 150?

3. Here are 24 squares. Circle 75% of the squares.



4. Han made 36 cookies for a bake sale. 50% of the cookies are chocolate chip and 50% are oatmeal raisin. How many oatmeal raisin cookies did Han bake?

5. Which is greater? Show or explain your thinking.

- A.** 100% of 25
- B.** 50% of 60
- C.** They are the same

6. Explain how you could mentally calculate 25% of any number.

7. Priya bought a pack of 12 pens. 50% of the pens are green, 25% are blue, and the rest are red.

- a** How many green pens did Priya buy?
- b** How many blue pens did Priya buy?
- c** How many red pens did Priya buy?

8. Jada and Diego each answer this question: 8 is what percent of 10?

- Jada says, “8 is 80% of 10.”
- Diego says, “8 is 20% of 10.”

Who is correct? Explain your thinking.

Additional Practice

ACC6.6.02

1. What percent of the diagram is shaded?



2. What percent of the diagram is shaded?



3. During batting practice, Bard hit 180 pitches and 25% of them were foul balls. How many foul balls did Bard hit? Show or explain your thinking.
4. 40% of the students in Clare's class earned an A on the last test. If there are 30 students in Clare's class, how many students earned an A? Show or explain your thinking.
5. Kiran wants to purchase a backpack for \$50. If the tax rate is 10%, how much will Kiran pay in tax?

6. Elena surveyed 250 students to find out about the favorite vegetables of students at her school. Complete the table to show either the number of students or the percent of students who said each vegetable was their favorite.

	Number of students	Percent of students (%)
Cucumbers	75	
Carrots	50	
Broccoli		10
Sweet potatoes	100	
Total	250	100

7. The recommended daily allowance of vitamin C for adults is 80 milligrams, and a 16-ounce smoothie says it contains 250% of the daily recommended allowance of vitamin C. How many milligrams of vitamin C are in the smoothie?
8. A large jug of milk contains 120 ounces of milk. A smaller jug of milk contains 65% as much milk as the large bottle. Jada says that there are 42 ounces of milk in the smaller bottle. Is Jada correct? Explain your thinking.

Additional Practice

ACC6.6.04

1. Select *all* the expressions that could be used to calculate 20% of 75.

- ☐ A. $\frac{75}{5}$
☐ B. $100 \cdot \frac{20}{75}$
☐ C. $\frac{20}{100} \cdot 75$
☐ D. $\frac{75}{100} \cdot 20$
☐ E. $\frac{100}{20} \cdot 75$

2. Select *all* the expressions that could be used to calculate 5% of 80.

- ☐ A. $\frac{5}{100} \cdot 88$
☐ B. $\frac{80}{100} \cdot 5$
☐ C. $\frac{88}{5} \cdot 100$
☐ D. $0.05 \cdot 88$
☐ E. $\frac{5}{88} \cdot 100$

3. Determine each missing value.

- a** What number is 12% of 125?

b 18 is 90% of what number?

c 120% of what number is 48?

4. Tyler and Andre scored 42% of their team's points at yesterday's basketball game. If their team scored 50 points, how many points did Tyler and Andre score?

5. A bakery sells 150 muffins on Tuesday. If 36% of the muffins were sold in the afternoon, how many muffins were sold in the afternoon?

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6. Lin spent 80% of her savings on a new phone. The phone cost \$450. How much money did Lin have saved?

7. An item is on sale for 30% off. The original price is \$42. What is the sale price of the item?

8. An athlete is training for a race and ran 32 kilometers this week. Next week, they plan to run 140% of that distance. How far do they plan to run next week?

Additional Practice

ACC6.6.05

1. Complete each percentage statement.

a 20% of 80 is

b 150% of 84 is

c 5% of 420 is

2. Complete each percentage statement.

a% of 20 is 15.

b% of 170 is 68.

c% of 20 is 25.

3. Complete each percentage statement.

a 14% of is 70.

b 225% of is 81.

c 10% of is 100.

4. On a sixth grade field trip, there are 5 chaperones for every 30 students. There are 175 people on the field trip.

a How many chaperones are on the field trip?

b How many sixth graders are on the field trip?

c What percent are chaperones? Round to the nearest whole percent.

d What percent are sixth graders? Round to the nearest whole percent.

- 5.** Andre conducts a survey of 1,450 people to find out about their favorite type of vacation. 56% of those surveyed prefer a beach vacation, 28% prefer a vacation in the mountains, and 16% prefer going to amusement parks for vacation.
- a** How many people prefer a beach vacation?
 - b** How many people prefer a mountain vacation?
 - c** How many people prefer going to an amusement park for vacation?
- 6.** A bakery made 1,175 bagels. 48% are onion bagels, 36% are sesame bagels, and 16% are raisin bagels.
- a** How many onion bagels were made?
 - b** How many sesame bagels were made?
 - c** How many raisin bagels were made?
- 7.** A bakery sells 12 corn muffins for every 4 lemon poppy seed muffins. On Saturday, the bakery sold 192 muffins.
- a** How many corn muffins were sold?
 - b** How many lemon poppy seed muffins were sold?
 - c** What percent of all the muffins sold were corn muffins?
 - d** What percent of all the muffins sold were lemon poppy seed muffins?
- 8.** Han says that 429 is 78% of 550. Is Han correct? Explain your thinking.

Additional Practice

ACC6.6.07

Problems 1–2: Students in two schools were asked whether they agreed with a rule that limits each student to at most one extra credit assignment per class, per semester. Use this information to solve each problem. Round your answers to the nearest whole percent.

1. Out of 355 students from school A, 175 agreed with the rule. What percent of students from school A agreed with the rule? Show or explain your thinking.

2. Out of 375 students from school B, 250 disagreed with the rule. What percent of students from school B disagreed with the rule? Show or explain your thinking.

3. An ice cream shop sold 380 ice cream cones. 35% of the cones were chocolate. How many chocolate ice cream cones did they sell?

4. 62 people watching a new movie were under 18. If 155 total people saw the movie, what percent of the audience was under 18?

5. Complete each percentage statement. Show your thinking.
 - a 85% of 360 is

 - b 17% of is 68.

 - c% of 36 is 63.

- 6.** Clare spent 50 minutes cooking dinner on Friday night.
- a** On Saturday night, it took Clare 130% as much time to cook dinner as it did on Friday night. How long did it take Clare to cook dinner on Saturday night?
 - b** On Sunday night, it took Clare 80% as much time to cook dinner as it did on Friday night. How long did it take Clare to cook dinner on Sunday night?
- 7.** An electronics store usually sells 550 video games each month. Determine each percentage.
- a** The store sold 99 games in the first week of the month. What percent of the sales for the month were in the first week?
 - b** In the first three weeks of the month, the store sold 462 games. What percent of the monthly sales were in the first three weeks?
 - c** During a different month, the store sold 660 games. What percent of the usual monthly sales is that?
- 8.** A pair of shoes normally costs \$80 and is on sale for 20% off. Tyler says, “The shoes cost \$20 less, which is \$60.” Is Tyler correct? Explain your thinking.

Additional Practice

ACC6.6.08

Problems 1–5: Fill in the blank to describe each percent increase or decrease as a percentage of the original amount.

1. A store manager decides on a 10% markdown on the retail price of the last pair of jeans in stock.

The price of a pair of jeans is % of the retail price.

2. Tickets to a pro basketball game are resold online with a 50% markup from their original price.

The price of tickets online to a pro basketball game is % of its original price.

3. The number of students enrolled at a local university dropped 14% from last year.

The number of students enrolled at a local university this year is % of the number of students that were enrolled last year.

4. This month, there was 30% more rain than last month.

The amount of rain this month is % of the amount last month.

5. This month, there were 15% fewer cloudy days than last month.

The number of cloudy days this month is % of the number of cloudy days last month.

6. At the start of the day, there was 12 inches of snow on the ground. Since then, 20% of the snow has melted. Which expression represents the inches of snow left on the ground?

- a. $0.2 \cdot 12$
- b. $0.8 \cdot 12$
- c. $(1 + 0.2) \cdot 12$
- d. $8 - 0.20$

Additional Practice

ACC6.6.09

1. There are 17% more girls in the eighth grade than in the seventh grade. If s is the number of girls in the seventh grade and e is the number of girls in the eighth grade, which equations represent the relationship between the number of girls in the seventh and eighth grades? Select *all* that apply.

☐ A. $e = 17 + s$

☐ B. $e = 1.17s$

☐ C. $e = (1 + 0.17)s$

☐ D. $e = s + 0.17s$

☐ E. $e = 17 + 0.17$

2. The retail price of a dress is marked down by 30%. If r is the retail price of the dress and s is the sale price after the markdown, which of the following equations do *not* represent the relationship between r and s ?

A. $s = (1 - 0.30)r$

B. $s = 1.30r$

C. $s = 0.7r$

D. $s = r - 0.30r$

3. Match each situation to the equation(s) that can be used to determine the unknown value.

- a. The population of geese in a northern city decreases by 68% from last season. If there are now 2,200 geese, how many geese were there last season?

..... $2200 = 1.68x$

..... $2200 = 0.68x$

..... $2200 = (1 - 0.68)x$

..... $2200 = 0.32x$

- b. The population of a certain type of beetle is now 2,200 after a 68% increase. What was the original population?

..... $2200 = x - 0.32x$

..... $2200 = (1 + 0.68)x$

..... $2200 = (1 - 0.32)x$

- c. There are 2,200 fish in a pond. This is after a 32% decrease in the fish population. What was the original population of fish in the pond?

..... $2200 = x + 0.68x$

..... $2200 = x - 0.68x$

4. Han's older brother works at the local department store. If Han shops there while his brother is working, he can get an 18% family discount on anything he buys. How much will a \$77 hoodie cost if Han uses his family discount? Show or explain your thinking.

5. A townhouse that was built several years ago has a current market value of \$130,000. If this represents an 18% increase in its market value since it was built, how much was it originally worth? Show or explain your thinking.

6. A summer camp has 200 children enrolled this year.
 - a If there were 125 children enrolled last year, what was the percent increase in enrollment from last year to this year?

 - b The camp director uses the percent increase from last year to predict the number of children that will enroll next year. How many children are predicted to enroll next year? Show or explain your thinking.

7. Kiran's grandfather gave him a vintage camera that is now worth \$3,000. This represents a 1,250% increase in its value from the time his grandfather bought the camera when he was a child. Kiran writes the equation $y = 12.50x$ to determine the price his grandfather paid for the camera, using y to represent the new value and x to represent the original value. Is his equation correct? Show or explain your thinking.

8. Explain why a percent increase can be greater than 100% but a percent decrease cannot be.

Additional Practice

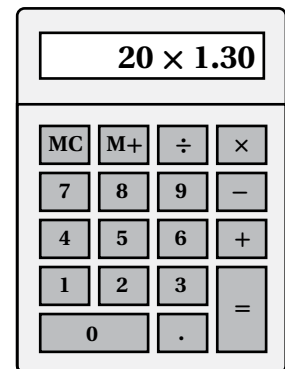
ACC6.6.10

1. The number of tomatoes in a garden decreased by 25% between last season and this season. This season, there are 63 tomatoes in the garden. What was the tomato population last season? Use a diagram if it helps with your thinking.
2. Next month, the price of a carton of strawberries at a grocery store will decrease by 30%. The price of a carton of strawberries this week is \$6.50. What will the price be next month?
 - A. \$8.45
 - B. \$6.75
 - C. \$4.55
 - D. \$3.45
3. A movie theater sells 8% more tickets this week than last week. If they sell 270 tickets this week, how many tickets did they sell last week? Use a diagram if it helps with your thinking.
4. The label on a carton of soup says that it has “20% less sodium than the original recipe.” If the soup has 2,000 milligrams of sodium in it, how much sodium is in the original recipe? Use a diagram if it helps with your thinking.

5. Kiran purchases two game systems so he can keep one and resell the other one when they sell out in stores. He finds a buyer who is willing to purchase one game system for \$900, which represents a 125% increase in the retail price of the game system. What was the price of the game system in the store? Use a diagram if it helps with your thinking.

6. An employee at a clothing store is asked to add a 130% markup on all \$20 shirts by their supervisor. The employee enters the expression shown into their calculator and then changes the prices of all the \$20 shirts to \$26.

- a What mistake did the employee make? Show or explain your thinking.
- b How much more should the shirts cost after a 130% markup?



7. Chloe noticed that a bag of lemons at the grocery store increased from \$2.00 to \$3.00 a bag. Chloe told the cashier that this represents a 150% increase. Without determining the actual percent increase, explain why Chloe's observation is not reasonable.

Additional Practice

ACC6.6.11

Problems 1–2: Determine the percent change. Use a tape diagram, if needed, to solve each problem. Show or explain your work. Round to the nearest tenth, if necessary.

1. Original Price: \$135
Markup Price: \$162

2. Original Price: \$162
Markdown Price: \$135

Problems 3–5: Jada plans to buy a pair of jeans on sale for \$45. When the sale is over, the jeans will return to their original retail price of \$75.

3. What is the percent change from the retail price to the sale price?
4. What is the percent change from the sale price back to the retail price?
5. Explain why the percent change in Problem 3 is *not* equal to the percent change in Problem 4.

Problems 6–7: A brand new car has a price tag of \$12,500.

6. If the same car is worth \$9,875 one year later, what is the percent change in the car's value from the previous year?
7. If the same car is worth \$8,295 one year after that, what is the percent change in the car's value from the previous year?

Problems 8–9: Han drinks 1 pint of water a day. He makes it a goal to increase his water intake.

- 8.** By what percentage does Han's water intake increase if he switches from a pint-sized water bottle to a quart-sized water bottle? **Hint:** 1 quart = 2 pints.
- 9.** By what percentage does Han's water intake increase if he switches from a quart-sized water bottle to a gallon-sized water bottle? **Hint:** 1 gallon = 4 quarts.

Problems 10–11: The population of a new middle school in its first year is 120 students.

- 10.** If the population of the school in its second year is 252 students, what is the percent change in the student population from the first year?
- 11.** If the population of the school in its third year is 363 students, what is the percent change in the student population from the second year?
- 12.** Last month, the price of windshield wiper fluid at a particular windshield wiper fluid station was \$1.29 per gallon. This month, it is \$2.70 per gallon. Without determining the actual percent increase, explain how you know the percent change in the price of windshield wiper fluid from last month to this month is greater than 100%.

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Additional Practice

ACC6.6.13

1. Han is charged \$11.70 sales tax on an item that costs \$195.00. Which of the following represents the tax rate in Han's state?

A. 0.06%

C. 16.7%

B. 6%

D. 167%

2. In Atlanta, GA, the sales tax rate is 8.9%. Complete the table.

Item	Price Before Tax (\$)	Sales Tax (\$)	Price Including Tax (\$)
Blouse	30		
Jeans	65		
Jacket	88		

3. Jada orders a takeout dinner for \$18 using a food delivery app. She has the option to tip the delivery driver 15%, 18%, or 20%.

- a Complete the table.

Percent of Tip (%)	15	18	20
Tip Amount (\$)			

- b If the sales tax rate is 8.5% of the cost without a tip, how much sales tax does Jada pay?
 - c If Jada chooses to tip 20%, what is the total cost of Jada's food delivery, including tax and tip?

4. Lin gets a pedicure for \$24 at her local nail salon.

- a** If a sales tax of \$2.10 is added to her bill, what is the sales tax rate in the city where the nail salon is located? Show or explain your thinking.
- b** If Lin tips her nail technician 12% of the cost of her pedicure before sales tax, how much does Lin pay in total at the nail salon, including the tax and tip? Show or explain your thinking.

5. Kiran and Priya eat at a restaurant together. Kiran orders an entree that costs \$38. Priya orders a different entree for \$24 and a dessert for \$14.

- a** Kiran says the cost of their meals, including tax, will be exactly the same. Do you agree with Kiran? Show or explain your thinking.
- b** Kiran and Priya are given a single receipt with both their orders on it. What is the tip amount if 18% gratuity is automatically added to their bill? Show or explain your thinking.
- c** The sales tax rate is 6.5% and is applied before tip. Kiran and Priya decide to split the cost of the meal evenly, including tax and tip, how much does each person pay? Show or explain your thinking.

Additional Practice

ACC6.6.14

Problems 1–2: A customer leaves a 20% tip on a \$25 meal.

1. Select the expression that represents the value of the tip.

- ☐ A. $20 \cdot 25$
- ☐ B. $25 + 2 \cdot 25$
- ☐ C. $1.20 \cdot 25$
- ☐ D. $\frac{20}{100} \cdot 25$

2. Select the expression that represents the total bill.

- ☐ A. $20 \cdot 25$
- ☐ B. $25 + 2 \cdot 25$
- ☐ C. $1.20 \cdot 25$
- ☐ D. $\frac{20}{100} \cdot 25$

Problems 3–4: Luciana works at a cafe. In an average 8-hour work day, she serves 15 customers, with an average bill of \$25 per customer. She typically receives a 15% tip on each bill and earns \$10.50 per hour.

3. How much money does Luciana earn on a typical day?

4. Let's say the typical tip increased to 20% of the bill. By what amount did Luciana's earnings increase by? Show or explain your thinking.

Problems 5–7: Here is some information about three local parks in a town.
Complete each sentence.

Park	Area (acres)
Sunset Cove	12
Cozy Corner	22
Morning Stroll	30

5. Cozy Corner is about% larger than Sunset Cove.
6. Morning Stroll is about% larger than Cozy Corner.
7. If the size of Sunset Cove increased by 150%, would it be larger than Morning Stroll?
Show or explain your thinking.

Problems 8–9: Jose is expanding the size of his house. He wants to add 10% to the square footage of his home. His home is currently 1,500 square feet.

8. Select the expression that represents the added square footage.
- ☐ A. $\frac{10}{100} \cdot 1500$
- ☐ B. $10 + 1.1 \cdot 1500$
- ☐ C. $1.10 \cdot 1500$
- ☐ D. $10 \cdot 1500$
9. Select the expression that represents the total square footage of Jose's home after the expansion is complete.
- ☐ A. $\frac{10}{100} \cdot 1500$
- ☐ B. $10 + 1.1 \cdot 1500$
- ☐ C. $1.10 \cdot 1500$
- ☐ D. $10 \cdot 1500$

Additional Practice

ACC6.6.15

1. Noah goes to the deli to order 2 pounds of turkey breast. When he receives his order, the weight is 1.95 pounds. Which expression represents the percent error?

A. $\frac{0.5}{1.95} \cdot 100$

C. $\frac{0.5}{2} \cdot 100$

B. $\frac{0.05}{1.95} \cdot 100$

D. $\frac{0.05}{2} \cdot 100$

2. A carnival has a game where players guess the number of marbles in a jar. One player guesses 225 marbles. The actual number of marbles is 299. What is the percent error, to the nearest tenth of a percent?

3. A tomato sauce recipe calls for 1 gram of sugar. When Diego weighs the sugar on his scale, he gets a weight of 1.03 grams. Determine the percent error of this measurement.

4. The thermostat in Andre's house gives a reading that is 4% below the actual temperature. If the actual temperature in Andre's house is 71°F, what is the reading on the thermostat, to the nearest tenth of a degree?

5. The water in the deep side of a pool has a depth of 80 inches.

a Mai gets in the pool and accurately estimates its depth to the nearest foot. What measurement did Mai get?

b By how many inches does Mai's measured depth differ from the actual depth of the pool?

c Determine the percent error of this measurement.

- 6.** Jada sends invitations for a barbecue she plans to have at the end of the school year. 68 people accept the invitation. On the day of the barbecue, 63 people attend.

a Which value is the actual value for the number of attendees to the barbecue? Which is the expected value? Show or explain your thinking.

b Determine the percent error of this measurement.

Problems 7–9: Han estimates his puppy weighs 40 pounds. During a visit to the vet, Han learns that his puppy actually weighs 44.6 pounds.

- 7.** What is the error, expressed as a percent of the puppy's actual weight?
- 8.** What is another value for the puppy's weight that would result in the same percent error? Show or explain your thinking.
- 9.** Is it possible to have 3 values that have the same percent error? Show or explain your thinking.

Additional Practice

4.10

1. A notebook costs \$8 before tax. A customer has a coupon for a 10% discount. Then a 5% sales tax is added. How much will the customer pay for the notebook?

Problems 2–3: Dalia needs to buy chili peppers for a recipe. At the grocery store, she finds red chili peppers that are available per ounce. The grocery store charges \$14 per ounce.

2. At the farmers market in the next town, Dalia remembers that the red chili peppers cost \$10 per ounce. By what percent did the price increase at the grocery store?

A. 10%
B. 25%

C. 40%
D. 60%
3. Dalia needs the chili peppers quickly for an upcoming dinner party, so she buys them at the grocery store. While checking out at the grocery store, she finds a coupon for a 25% discount. Then a 6% sales tax is added. How much will Dalia pay for the chili peppers if she buys 1 ounce, 3 ounces, and 4 ounces? Fill the missing values in the table.

Red chili peppers (in ounces)	Cost (in \$)	Amount of coupon discount (in \$)	Amount of sales tax (in \$)	Amount Dalia will pay
1	14	3.50	0.63	
3	42		1.89	
4		14		

Problems 4–6: A clothing store allows customers to use multiple coupons when checking out. You want to buy a jacket that costs \$57. Suppose you have a \$7 off coupon and a 15% off coupon. The cashier will calculate the new price of the jacket after each coupon is used.

4. You use the \$7 off coupon first and then the 15% off coupon. What is the total price of the jacket after you hand the cashier the coupons in this order?
5. Your friend, Jackie, buys the same jacket. However, Jackie hands the cashier the 15% off coupon first. Then, she hands the cashier the \$7 off coupon. How much does Jackie pay for the jacket?
6. Does the order you use the coupons make a difference? Explain your thinking.

Problems 7–9: An art shop in town has a 5% sales tax.

7. A pack of pencils costs \$6.80 before tax. How much does it cost including tax?
8. A paintbrush set costs \$31.50 after tax. How much did it cost before tax?
9. A sketchbook costs \$18.00 after tax. How much did it cost before tax?

Additional Practice

ACC6.6.16

- Juniper is painting her kitchen with a shade of paint called Sunset Yellow. To make this shade, Juniper needs to mix 6 quarts of yellow paint with $\frac{1}{3}$ cups of orange paint. How much yellow paint should be mixed with 2 cups of orange paint to make Sunset Yellow?

Problems 2–3: A local organic spice store offers:

$3\frac{1}{2}$ ounces of basil for \$21.00 $\frac{5}{8}$ ounces of rosemary for \$3.50 $1\frac{1}{2}$ ounces of oregano for \$9.90

- Which spice is the least expensive per ounce? Show or explain your thinking.

- Which spice is the most expensive per ounce?

Problems 4–5: Emilia wants to buy stickers at a stationary store. The stickers are sold by the sheet.

- Emilia has \$60 to spend on stickers. How many sheets of stickers can she buy? Fill the missing value into the table.

Number of Sheets	Total Cost (\$)
2	6
	60

- Which equation represents the relationship between the total cost, c , and the number of sticker sheets, s ?

A. $s = \frac{1}{3}c$

B. $s = 3c$

C. $c = \frac{1}{3}s$

D. $c = 2s$

Problems 6–8: A tortoise is moving away from the beach at a constant rate. This table shows the distance the tortoise is from the beach at certain times.

Distance (ft)	Time (min)
0	0
1	$1\frac{2}{3}$

6. How many minutes does it take for the tortoise to reach a distance of 5 feet from the beach?
7. How far will the tortoise be from the beach after 8 minutes?
8. Select *all* the equations that represent the relationship between the distance in feet, d , and time in minutes, t .
- ☐ A. $t = \frac{5}{3}d$
- ☐ B. $t = \frac{3}{5}d$
- ☐ C. $d = \frac{5}{3}t$
- ☐ D. $d = \frac{3}{5}t$

Additional Practice

ACC6.6.17

1. Select *all* the ratios that are equivalent to 3 : 4.

☐ A. 6 : 8

☐ B. 1.5 : 2

☐ C. 4 : 5

☐ D. 9 : 12

☐ E. 15 : 16

2. On a map, the library is 1.5 inches from Cordelia's house. The map has a scale of 1 inch to 10 miles. How far apart, in inches, would Cordelia's house be from the library on a map that has a scale of 1 inch to 60 miles?

3. What is 30% of 180?

4. What is 180% of 60?

5. Complete the table to represent $y = \frac{1}{3}x$.

x	y
12	
	$6\frac{1}{3}$
24	

Problems 6–7: To make a specific color of purple paint, a painter mixes $\frac{1}{3}$ of a gallon of red paint with $\frac{3}{5}$ of a gallon of blue paint.

6. How many gallons of red paint are needed to mix with 3 gallons of blue paint to make this specific shade of purple?
7. How many gallons of blue paint are needed to mix with 10 gallons of red paint to make the same shade?
- A. 12
- B. 14
- C. 16
- D. 18
8. A brownie recipe calls for $\frac{1}{4}$ teaspoon of baking powder and 1 cup of flour. Complete the table to show how much baking powder and flour is needed for bigger and smaller batches of the brownie recipe.

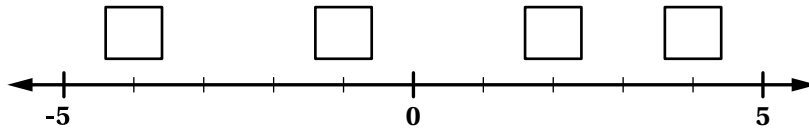
Baking Powder (teaspoons)	Flour (cups)
$\frac{1}{4}$	1
$\frac{3}{4}$	3
$1\frac{1}{2}$	6
	8
	$10\frac{1}{2}$

Additional Practice

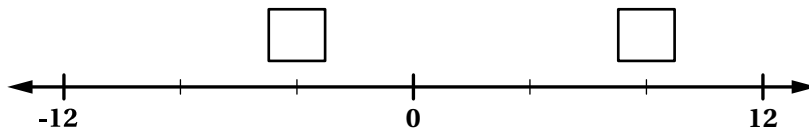
ACC6.7.01

Problems 1–4: Fill in the blanks on the number lines.

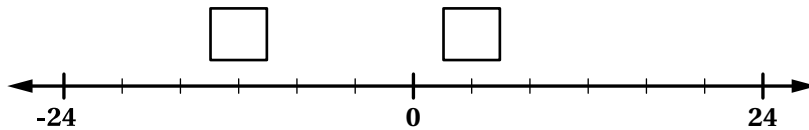
1.



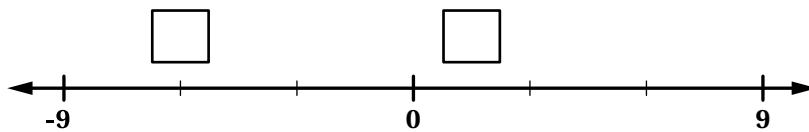
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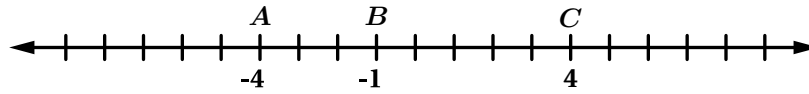
3.



4.



Problems 5–8: Here is a number line.



5. Describe where you would plot 50 on the number line.

6. Point *D* is 3 units to the left of point *A*. Plot point *D*.

7. Point *E* is at 0. Where is Point *E* located?

- A. To the left of point *A*
- B. Between points *A* and *B*
- C. Between points *B* and *C*
- D. To the right of point *C*

8. What locations are 2 units away from point *B*?

- A. -3 and 1
- B. -4 and 2
- C. -5 and -2
- D. -6 and -5

Additional Practice

ACC6.7.02

1. For each number, name its opposite.

a 0

b 11

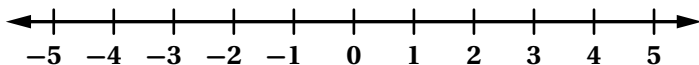
c $-1\frac{1}{3}$

d -3.125

e 9.15

f $\frac{3}{4}$

2. Plot and label each point on the number line.



a Point *A* is located at the opposite of 0.

b Point *B* is located at the opposite of 2.5.

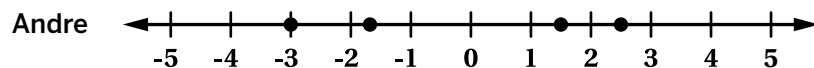
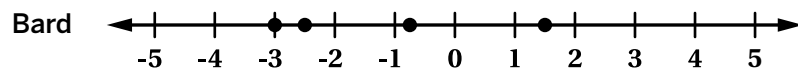
c Point *C* is located at the opposite of -4 .

d Point *D* is located at -1 .

e Point *E* is located at 3.5.

f Point *F* is located at $-\frac{5}{3}$.

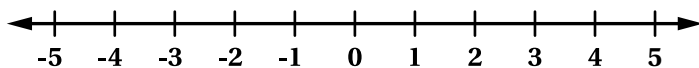
Problems 3–5: Bard and Andre each plotted the numbers $-\frac{3}{4}$, $1\frac{1}{2}$, -3 , and 2.5 on the number lines shown. They each made mistakes.



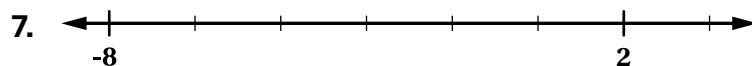
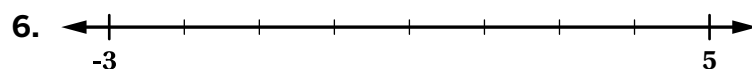
3. Which number did Bard plot incorrectly? What question would you ask Bard to help understand the mistake?

4. Which number did Andre plot incorrectly? What question would you ask Andre to help understand the mistake?

5. Plot the numbers $-\frac{3}{4}$, $1\frac{1}{2}$, -3 , and 2.5 in the correct location on the number line shown.



Problems 6–7: Plot and label where zero is located on each number line.



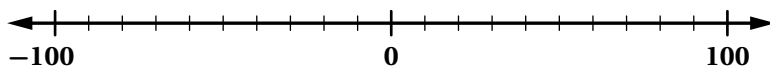
Additional Practice

ACC6.7.03

1. Write a positive or negative integer that represents each elevation.

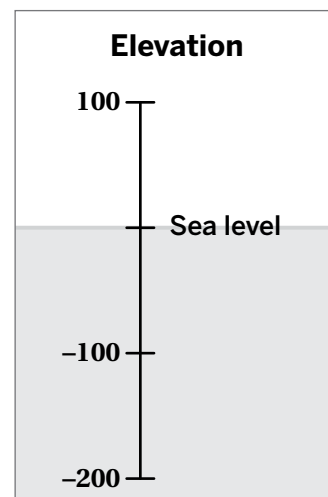
- a The Dead Sea has an elevation of 430 meters below sea level.
- b Santa Fe, New Mexico, has an elevation of 2,194 meters above sea level.
- c Indio, California, has an elevation of 6 meters below sea level.
- d Lake Eyre, Australia, has an elevation of 15 meters below sea level.
- e Flagstaff, Arizona, has an elevation of 2,106 meters above sea level.

2. Baku, Azerbaijan, has an elevation of 92 feet below sea level. Suva, Fiji, has an elevation at sea level. Phnom Penh, Cambodia has an elevation of 39 feet above sea level. Plot and label each elevation on the number line.



3. Three humpback whales move in the ocean.

- a A humpback whale is at the surface of the ocean to breathe. What is the whale's elevation?
- b A whale starts at the surface, then dives down 180 feet to feed. What is the whale's elevation now?
- c A whale starts at the surface, then breaches (leaps) 10 feet into the air. What is the whale's elevation as it leaps?
- d Plot all of the elevations on the number line.



4. Complete these problems about sea level and elevation.

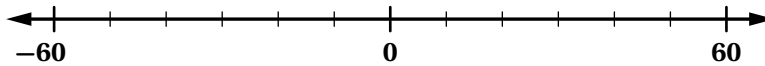
- a Which elevation is closer to sea level: -20 meters or 25 meters?
- b Which elevation is closer to sea level: 10 meters or -20 meters?

5. Complete the table for each elevation with its relation to sea level.

Elevation	Relation to Sea Level (Above/Below/At)
-26 m	
0 yd	
15 ft	

6. These statements describe the movements of a hiker on a trail. Each statement starts from the hiker's elevation in the previous statement.

- a A hiker starts at the base of the trail which is at sea level. What is the hiker's elevation?
- b The hiker then walks up 24 meters to a small summit. What is the hiker's elevation now?
- c Plot and label the two elevations on the number line.



7. Clare claims that a city with an elevation of -15 feet is closer to sea level than a city with an elevation of 12 feet. Is Clare correct? Explain your thinking.

Additional Practice

ACC6.7.04

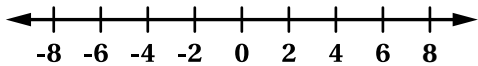
1. Complete each statement with a number that makes it true. Use the number line if it helps with your thinking.

a $\square > 3$

b $\square > -3$

c $-3 > \square$

d $-3 < \square$



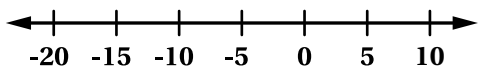
2. Complete each statement with the symbol $<$, $>$, or $=$. Use the number line if it helps with your thinking.

a $1 \square -3$

b $-3 \square 3$

c $-13 \square -11$

d $-11 \square -11.5$



Problems 3–6: Circle whether each statement is *true* or *false*.

3. -2.7 is to the left of -1.9 on the number line. True False

4. -3.2 is greater than -2.1 . True False

5. $-\frac{5}{9} > -\frac{8}{9}$ True False

6. Choose one statement and explain your thinking.

7. Here are five numbers: $-\frac{3}{4}$, -2 , 1 , $\frac{3}{2}$, 2

These numbers are plotted on a horizontal number line. Which statement about the locations of the numbers is true?

- A. 2 is the farthest to the right, and $-\frac{3}{4}$ is farthest to the left.
 B. 2 is the farthest to the right, and -2 is farthest to the left.
 C. 1 is the farthest to the right, and $-\frac{3}{4}$ is farthest to the left.
 D. $-\frac{3}{4}$ is the farthest to the right, and 2 is farthest to the left.

Problems 8–9: Order each set of numbers from *least* to *greatest*.

8.

-2	2.1	-2.4	2.4	$\frac{2}{3}$
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Least Greatest

9.

$-\frac{4}{9}$	$-\frac{9}{4}$	-4.9	-9.4	4.9
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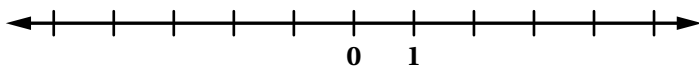
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Least Greatest

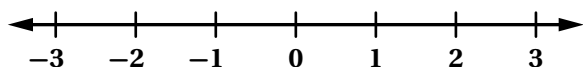
Additional Practice

ACC6.7.05

1. Plot and label all numbers that are 3 units away from 0.



2. Plot and label all numbers with an absolute value of $\frac{3}{2}$.



3. Complete each statement with the symbol $<$, $>$, or $=$.

a $2.1 \square -4.5$

b $|2.1| \square |-4.5|$

c $-5 \square 4.5$

d $|-5| \square |4.5|$

e $-4.5 \square \frac{9}{2}$

f $|-4.5| \square \left| \frac{9}{2} \right|$

g $|-2.1| \square |-3.2|$

h $|-2.1| \square -3.2$

4. Order these values from *least* to *greatest*.

$|3.4|$, $|-2|$, 0 , 2.6 , -1

5. Determine the absolute value of each number.

a $\left| -\frac{11}{12} \right|$

b $|-8|$

c $\left| 1\frac{3}{4} \right|$

d $|3.9|$

6. Shawn claims that the expressions 3.8 and $|-3.8|$ do not have the same value because $|-3.8|$ includes a negative symbol and an absolute value symbol. Is Shawn correct? Explain your thinking.

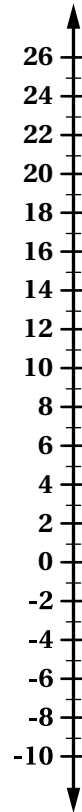
Additional Practice

ACC6.7.06

Problems 1–5: Melanie is studying different cities' elevations using a vertical number line. She wants to compare the heights of cities above and below sea level. Here are the elevations of four cities:

- Miami, FL: 3 feet above sea level
- New Orleans, LA: 6 feet below sea level
- Amsterdam, Netherlands: 7 feet below sea level
- Hamburg, Germany: 24 feet above sea level

1. Which city has the lowest height in relation to sea level?
2. The height of Venice, Italy is at sea level. Which of these 4 cities has the greatest difference between their height and Venice's?
3. How much higher is Miami than New Orleans?
4. How much higher is Hamburg than Amsterdam?
5. How much higher is Hamburg than New Orleans?
6. Jericho is the lowest city on Earth, with an elevation of 250 meters below sea level. Jerusalem is 400 meters above sea level. What is the difference between their elevations? Show or explain your thinking.



Additional Practice

ACC6.7.07

Problems 1–4: Determine the value of each expression.

1. $6 + -2$

2. $-6 + 2$

3. $-6 - 2$

4. $-6 - (-2)$

5. The temperature was 16°F and then dropped 12 degrees. What was the final temperature?

6. The temperature was -16°F and then it dropped 12 degrees. What was the temperature?

7. The temperature was -16°F and then rose to 12 degrees. What was the change in temperature?

8. A swimmer was 6 feet underwater. Then he swam 4 feet deeper. Katrina wrote the expression $-6 - 4$. Rodney wrote the expression $-6 + (-4)$. Explain why both Katrina and Rodney are correct.

Problems 9–10: The table shows eight expressions.

9. Determine the value of each expression.

	Expression	Value
Expression 1	$2 + 4 - 6$	
Expression 2	$2 + 4 - 6 + 8$	
Expression 3	$2 + 4 - 6 + 8 - 10$	
Expression 4	$2 + 4 - 6 + 8 - 10 + 12$	
Expression 5	$2 + 4 - 6 + 8 - 10 + 12 - 14$	
Expression 6	$2 + 4 - 6 + 8 - 10 + 12 - 14 + 16$	
Expression 7	$2 + 4 - 6 + 8 - 10 + 12 - 14 + 16 - 18$	
Expression 8	$2 + 4 - 6 + 8 - 10 + 12 - 14 + 16 - 18 + 20$	

10. What is the value of the 9th expression? The 10th expression? The 20th expression?
Show or explain your thinking.

Additional Practice

ACC6.7.08

Problems 1–3: Determine the value of the variable that makes each equation true.

1. $12 + a = 5$

2. $-4.5 + b = 6.5$

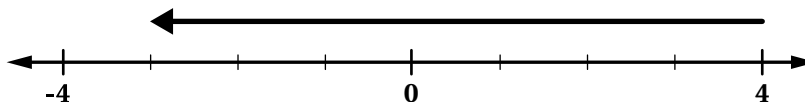
3. $c + 5.1 = 2.8$

Problems 4–5: Evaluate each expression.

4. $6 - 8$

5. $-6 - 8$

6. Circle the equation that is represented by this number line.



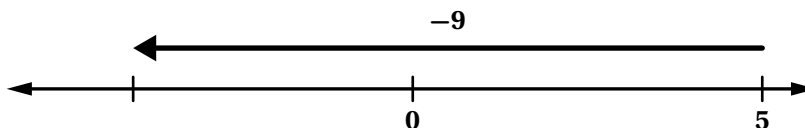
A. $x + (-3) = 4$

B. $4 + x = -3$

C. $-3 + x = 4$

D. $4 + (-3) = x$

7. Circle the equation that is represented by this number line.



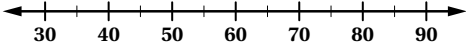
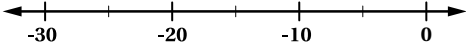
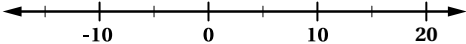
A. $x + (-9) = 5$

B. $-9 + x = 5$

C. $5 + x = -9$

D. $5 + (-9) = x$

Problems 8–10: Draw an arrow on the number line to represent each situation.

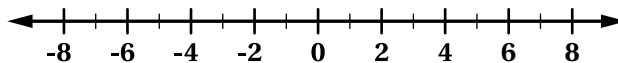
Situation	Number Line	Equation
8. The temperature was 60°F and then fell 15°F.		$60 + (-15) = x$
9. The temperature was −20°F and then rose 5°F.		$-20 + 5 = x$
10. At sunrise, the temperature was −10°F. At noon, the temperature is 15°F. By how much did the temperature rise?		$-10 + x = 15$

Additional Practice

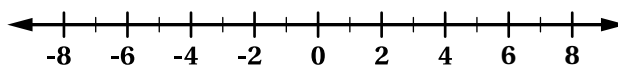
ACC6.7.09

Problems 1–3: Use the number lines to determine the value of each expression.

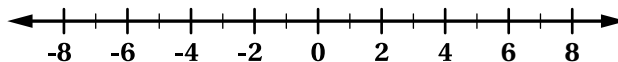
1. $2 - 6$



2. $(-1) + (-4)$



3. $-4 - (-8)$



Problems 4–6: Complete the tables.

4.

Expression	Value
$4 + (-5)$	
$-5 + 3$	
$-3 + (-10)$	
$-1.5 + (-3.7)$	

5.

Expression	Value
$-5 + 4$	
$3 + (-5)$	
$(-10) + (-3)$	
$-3.7 + (-1.5)$	

6. Use your work from the previous tables to describe any patterns you notice.

Problems 7–9: Complete the tables.

7.

Expression	Value
$4 - 5$	
$-5 - 3$	
$-3 - (-10)$	
$-1.5 - (-3.7)$	

8.

Expression	Value
$5 - 4$	
$3 - (-5)$	
$(-10) - (-3)$	
$-3.7 - (-1.5)$	

9. Use your work from the previous tables to describe any patterns you notice.

10. The expression $x + y$ equals -3 . If y is greater than 5, what are two possible values for x ?

Additional Practice

ACC6.7.10

Problems 1–2: Order the expressions from *least* to *greatest* value.

1.

$7 + (-5)$	$-7 + 5$	$7 + 5$	$-7 + (-5)$

Least **Greatest**

2.

$7 - (-5)$	$-7 - 5$	$7 - 5$	$-7 - (-5)$

Least **Greatest**

3. Select *all* expressions that will result in a negative value.

☐ A. $105 + (-74)$

☐ B. $-266 + (-87)$

☐ C. $141 - (-74)$

☐ D. $-130 - (-43)$

☐ E. $-100 - (-101)$

☐ F. $120 + (-121)$

4. Explain your thinking for Problem 3.

5. A bird is 15.2 feet above the ocean and dives to catch a fish 3.5 feet below the surface of the water. Which expression shows the total vertical distance the bird traveled?

A. $15.2 - 3.5$

B. $15.2 - (-3.5)$

C. $15.2 + (-3.5)$

D. $-3.5 + (-15.2)$

Problems 6–9: Determine the value of the variable that makes each equation true. Show your thinking.

6. $24 + w = -24.6$

7. $-12 - x = 10.6$

8. $y = \left(-\frac{1}{4}\right) + \frac{3}{8}$

9. $z + 8.9 = -16$

- 10.** One of the coldest places on Earth is Denali, Alaska, where it can get as low as -73.8°C . One of the hottest locations on Earth is Death Valley, California, where it can get as high as 56.7°C .

- a** Write two different expressions to determine the difference in temperatures between Denali and Death Valley.

- b** Determine the difference in temperatures between these two locations.

Additional Practice

ACC6.7.11

Problems 1–3: Determine the value of the variable that makes each equation true.

1. $6 \cdot a = -18$

2. $-6 \cdot (3) = b$

3. $-6 \cdot c = 18$

Problems 4–6: A weather station on top of a mountain reports that the temperature is currently -10°F and has been decreasing at a constant rate of 2 degrees per hour.

4. What will the temperature be in 5 hours?

5. What was the temperature 2 hours ago?

6. What was the temperature 4 hours ago?

Problems 7–9: Mark if each equation is *true* or *false*. If the equation is false, change one value to make it true, and write the revised equation in the last column of the table.

		True	False	Revised Equation
7.	$(-4) \cdot (-5) = -20$			
8.	$8 \cdot (-3) = 24$			
9.	$(-6) \cdot (-1) = 6$			

10. Complete the table for each action the submarine takes when starting at 0 units. Write an expression for each action, the direction the submarine will move, and its final value.

Action	Expression	Direction	Final Value
Adding 2 groups of 4 floats			
Removing 3 groups of 2 floats			
Adding 4 groups of 3 anchors			
Removing 3 groups of 1.5 anchors			

Additional Practice

ACC6.7.12

1. Match each expression to its value.

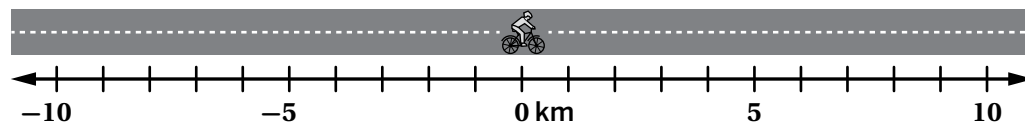
- | | |
|--------------------|-----------|
| a. $4 \cdot 3$ | _____ -5 |
| b. $2 \cdot (-2)$ | _____ -12 |
| c. $4 \cdot (-3)$ | _____ 5 |
| d. $5 \cdot (-1)$ | _____ 12 |
| e. $-1 \cdot (-5)$ | _____ -4 |

2. Determine if each expression has a positive or negative product.

$2 \cdot (-2)$	$2 \cdot 2$	$\frac{1}{2} \cdot \left(-\frac{1}{4}\right)$	$3.5 \cdot 43$	$0.001 \cdot \frac{1}{10}$
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Positive	Negative

Problems 3–4: The cyclist at kilometer 0 is moving at a steady speed.



3. Complete the table to determine the cyclist's position at various times.

Time (minutes)	-6	-4	-2	0	2	4
Position (km)			-4	0	4	

4. Determine the position of the cyclist at each time.

- a** 5 minutes **b** -10 minutes

5. Determine the number that belongs in each box to make the equation true.

a $\square \cdot (-2) = -24$

b $\square \cdot (-8) = -24$

c $2 \cdot \square = -12$

d $4 \cdot \square = -24$

6. Complete the missing expressions and values in the table.

Expression as a product	Value
$2 \cdot (-3)$	
	-1
$5 \cdot 0.4$	
	-7
$6 \cdot (-0.1)$	

7. Tyler and Bard are cycling on the boardwalk at the same time. When they pass the pier, Bard is cycling at a speed of 15 feet per second and Tyler is cycling at a speed of 10 feet per second.

a The pier is at position 0. How many feet past the pier will each person be in 5 seconds? Show or explain your thinking.

b How many feet in front of Bard was Tyler 8 seconds before arriving at the pier? Show or explain your thinking.

8. Han says that $-(a \cdot b)$ will always have the same product as $a \cdot (-b)$. Is Han correct? Show or explain your thinking.

Additional Practice

ACC6.7.13

1. Select *all* the expressions that have a negative value.

☐ A. $-\frac{12}{10}$

☐ B. $-\frac{12}{10}$

☐ C. $\frac{12}{10}$

☐ D. $-\frac{12}{-10}$

☐ E. $\frac{12}{-10}$

Problems 2–3: Determine the value of x that makes each equation true.

2. $-5x = -30$

3. $\frac{x}{-2.5} = 15$

Problems 4–5: The price of a gaming system declined \$83.88 over the course of the year (12 months).

4. If the change was constant, how much did the price change each month? Make sure you show whether the charge was negative or positive. Show or explain your thinking.

5. How much did the price change in the first 5 months?

6. Select *all* the values that are equivalent to $-\frac{15}{8}$.

☐ A. $-1\frac{7}{8}$

☐ B. $1\frac{7}{8}$

☐ C. $\frac{-15}{8}$

☐ D. $-\frac{15}{-8}$

☐ E. $\frac{15}{-8}$

7. Order these expressions from *least* to *greatest*.

$-12 \div (-4)$	$-12 \div \left(-\frac{1}{4}\right)$	$12 \div (-4)$	$-12 \div \left(\frac{1}{4}\right)$

Least **Greatest**

Problems 8–11: Determine the missing value in each equation.

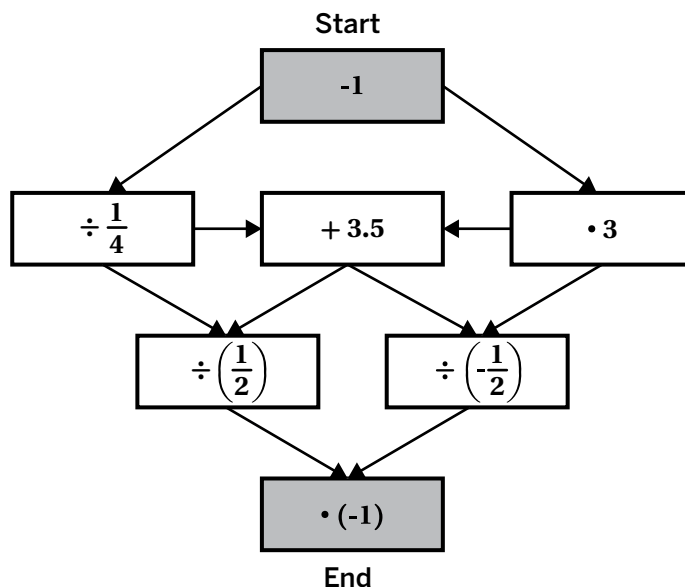
8. $-45 \div \dots = -9$

9. $-21 \div \dots = 3$

10. $\dots \div 8 = -9$

11. $\dots \div 2 = -\frac{1}{4}$

12. Move from box to box performing the indicated operation as you reach each new box. You must follow the direction of the arrows. Complete at least two paths. What is the lowest value you were able to determine?



Additional Practice

ACC6.7.15

Problems 1–4: Determine the value of the variable that makes each equation true.

1. $-12 + a = -16$

2. $-12 - 16 = b$

3. $-4c = -12$

4. $\frac{d}{-4} = 12$

5. Circle the expression that has the *lesser* value.

A. $(-12) - (-4)$

B. $(-4) - (-12)$

C. They have the same value

Problems 6–7: Let $x = 3$, $y = -4$, and $z = -3$.

6. Order these expressions from *least* to *greatest*.

$x - z$	$x - 2y$	$x \cdot y$	xyz

Least

Greatest

7. Would your order be different if the value of x was -3 instead? Explain your thinking.

8. For the expressions $\frac{a}{b}$ and $a + b$, choose values for a and b so that $\frac{a}{b}$ is negative and $a + b$ is positive.

Problems 9–10: Here are 2 sets of values for c and d .

9. Complete the table.

c	d	$c + d$	$c - d$	$-c + d$
$-\frac{2}{3}$	$2\frac{5}{6}$			
$\frac{2}{3}$	$-2\frac{5}{6}$			

10. Describe any patterns you notice.

Additional Practice

ACC6.7.16

Problems 1–4: Determine the value of the variable that makes each equation true.

1. $4 \cdot (-3.4) = a$

2. $-10b = 40$

3. $-4 - \frac{2}{5} = c$

4. $\frac{d}{4} = -8.2$

5. For each equation, select an operation (+, −, •, ÷) to make the equation true.

a $-18 \dots\dots 2 = -9$

b $9 \dots\dots \left(-\frac{2}{3}\right) = -6$

c $-9 \dots\dots \left(-\frac{3}{4}\right) = 12$

d $11 \dots\dots (-15) = -4$

e $-12 \dots\dots (-15) = 3$

f $-12 \dots\dots (-10) = 120$

6. A submarine starts at the ocean surface and descends toward the ocean floor at a rate of 25 meters per minute for 30 minutes. Select the expression that could represent the depth of the submarine after its descent.

A. $-25 \div 30$

B. $-25 \cdot 30$

C. $-25 + 30$

D. $-25 - 30$

7. Each table shows deposits and withdrawals for different bank accounts. Select the bank account that currently has the highest balance. Assume each account started with the same amount.

A.

Andre
−\$30.50
\$19.50
\$42.40

B.

Elena
\$45.50
−\$14.00
\$14.00

C.

Clare
−\$23.70
\$95.70
−\$60.40

D.

Tyler
\$25.50
−\$70.30
\$50.40

8. Match each situation to the equation that could represent it.

..... The temperature is decreasing at a rate of 1.5°C per second. How long will it take for the temperature to be 6°C colder than it is now?

a. $1.5c = -6$

..... A penguin descended below the surface of the ocean. After 1.5 seconds, the penguin was 6 feet below the surface. At what rate was the penguin diving?

b. $-6 + 1.5 = c$

..... A seal was swimming at -6 meters below sea level and then swam 1.5 meters toward the surface. What is the seal's elevation now?

c. $-1.5c = -6$

..... The temperature was -1.5°C and changed to -6°C . What was the change in temperature?

d. $-1.5 + c = -6$

Additional Practice

ACC6.7.17

Problems 1–2: Here is a coordinate plane.

1. Plot and label the following four points:

$A(1, 1)$, $B(-4, 1)$, $C(-1, -1)$, $D(1, -4)$.

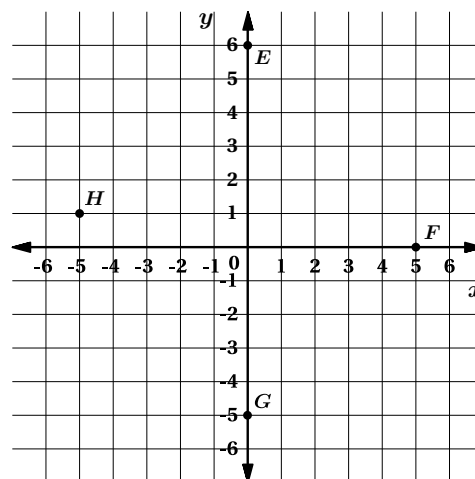
2. Write the coordinates for each point.

a Point E

b Point F

c Point G

d Point H



Problems 3–4: Here is a coordinate plane.

3. Plot and label the following four points:

$A(0, 7)$, $B(-7, 0)$, $C(-4, -3)$, $D(4, 3)$.

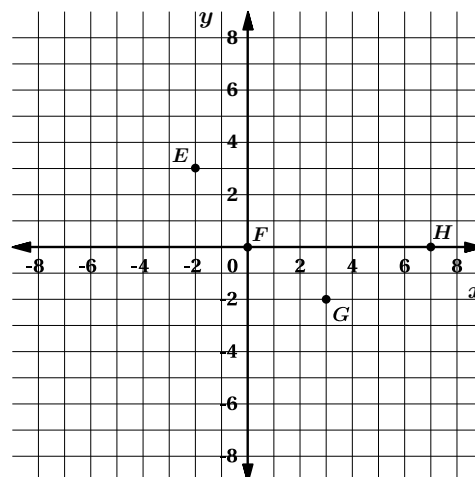
4. Write the coordinates for each point.

a Point E

b Point F

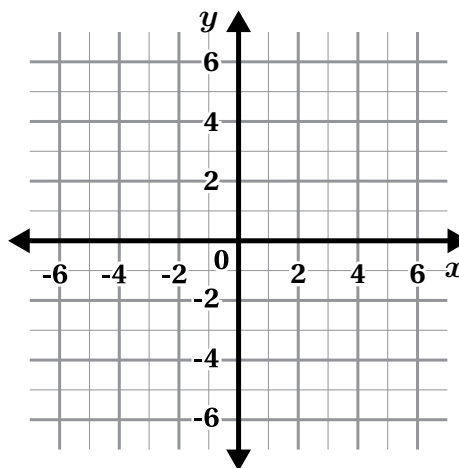
c Point G

d Point H



5. When these three ordered pairs are plotted, the points fall on the same line: $(-3, -3)$, $(-3, 0)$, $(-3, 1)$. Use the coordinate plane if it helps with your thinking.

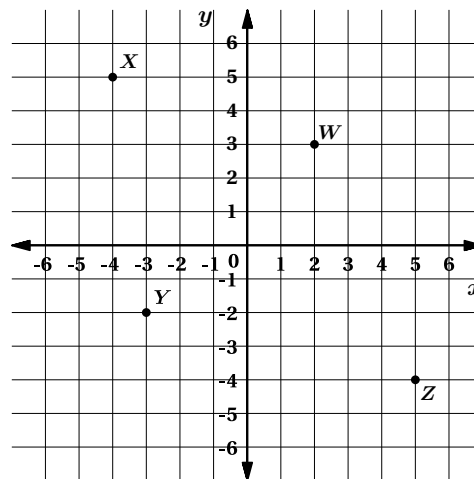
- a Is the line vertical or horizontal? Explain your thinking.
- b Write the ordered pairs for two other points that also lie on this same line.



6. Here is a coordinate plane.

- a Complete the table to show the coordinates and quadrant for each point.

Point	Coordinates	Quadrant
W		
X		
Y		
Z		



- b Point V is located at $(0, 5)$. On what axis does point V lie on?
- c For all of the points in Quadrant 3, are both coordinates *positive* or *negative*?
7. Priya says that if both coordinates of a point are positive, the point can be located in either Quadrant 1 or Quadrant 2. Is Priya correct? Explain your thinking.

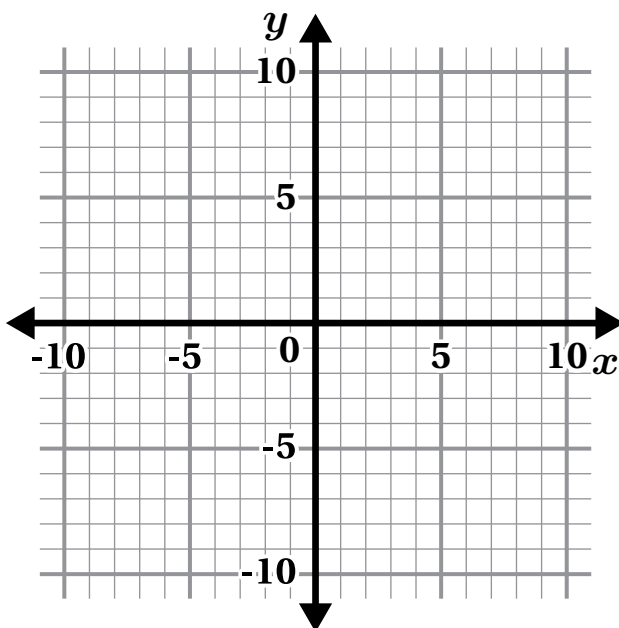
Additional Practice

ACC6.7.18

Problems 1–2: Here is a graph.

- Plot and connect each point in order.

Coordinates
$(-6, 5)$
$(0, 1)$
$(6, 5)$
$(6, -5)$
$(0, -1)$
$(-6, 5)$

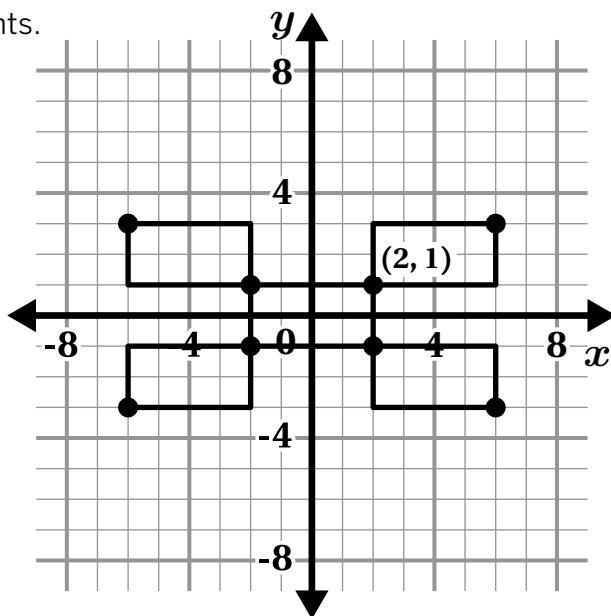


- Describe your strategy for plotting the point $(-6, 5)$ in Problem 1.

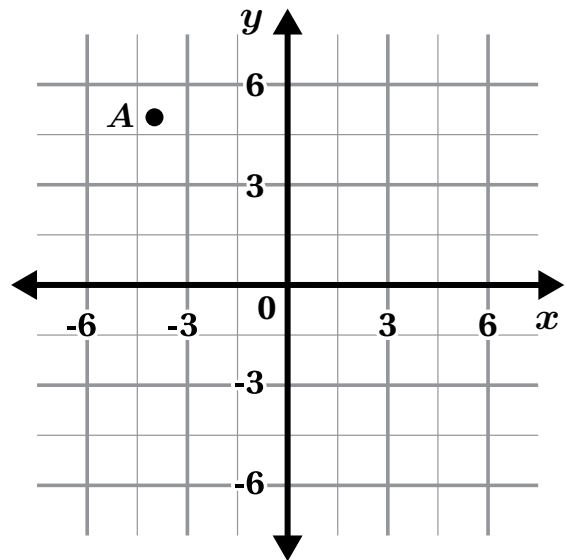
Problems 3–4: Here are identical rectangles on a coordinate plane. The origin is in the center of the middle rectangle.

- Label the coordinates of the remaining points.

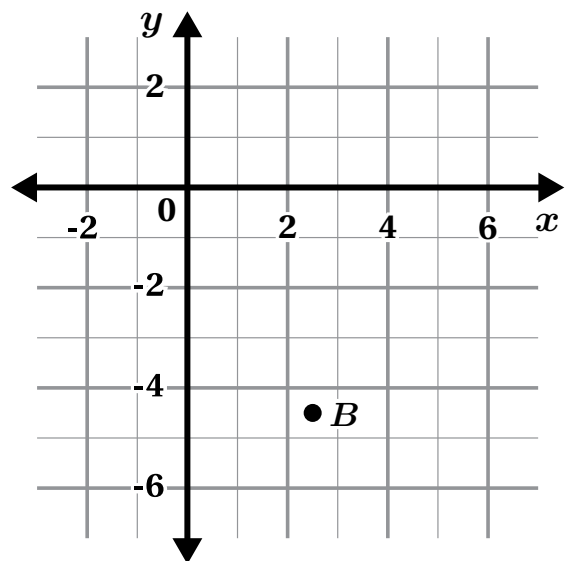
- What patterns do you notice?



5. What is the value of the x -coordinate of point A ? Your answer should be an integer.



6. What is the value of the y -coordinate of point B ? Your answer should be a decimal rounded to the nearest 0.5.



7. Point P is located at $(5, n)$ on the coordinate plane. Point Q is located at $(-5, n)$. What do you know about points P and Q ?
8. Point M is located at $(q, 1)$ on the coordinate plane. Point L is located at $(q, -1)$. What do you know about points M and L ?

Additional Practice

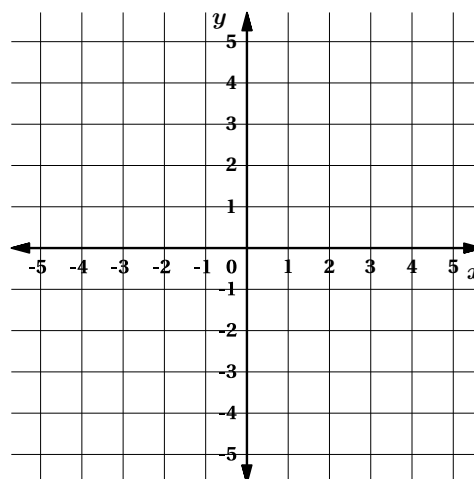
ACC6.7.19

1. The vertices of a rectangle are $(-2, 2)$, $(-2, -4)$, $(2, -4)$, and $(2, 2)$.

- What are the length and width of the rectangle?
- Determine the perimeter of the rectangle.
- Calculate the area of the rectangle.

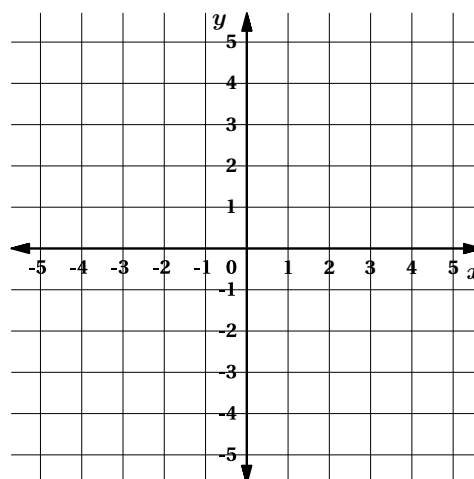
2. Here is a coordinate plane.

- Sketch a square with a perimeter of 24 units and one vertex located at the point $(4, 4)$.
- Write the coordinates of the other vertices.
- Calculate the area of the square.



3. Here is a coordinate plane.

- Sketch a square with a perimeter of 20 units and one vertex located at the point $(-1, 1)$.
- What are the coordinates of the other vertices?
- Calculate the area of the square.



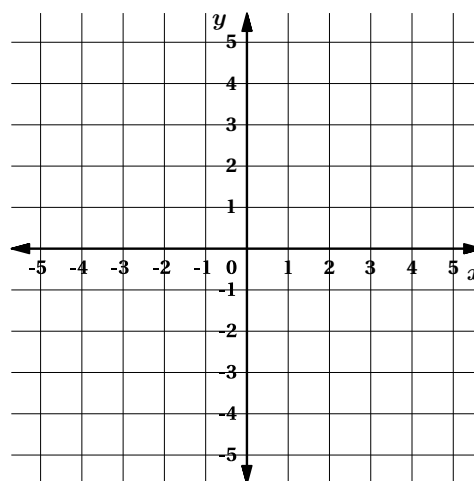
Problems 4–5: Here is a coordinate plane.

- 4.** Plot each point in the order listed: $(3, 3)$, $(3, -4)$, $(-4, -4)$, $(-4, 1)$, $(1, 1)$, $(1, 3)$.

- 5.** Connect the points to create a polygon.

a What is the perimeter of the polygon?

b What is the area of the polygon?



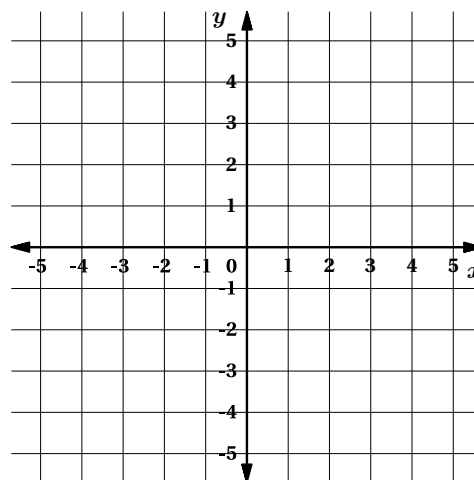
Problems 6–7: Here is a coordinate plane.

- 6.** Plot each point in the order listed. Label the polygons A and B.

Polygon A: $(-5, 5)$, $(-4, 5)$, $(-4, -1)$, $(1, -1)$, $(1, -4)$, $(-5, -4)$

Polygon B: $(-3, 5)$, $(-3, 4)$, $(2, 4)$, $(2, -4)$, $(5, -4)$, $(5, 5)$

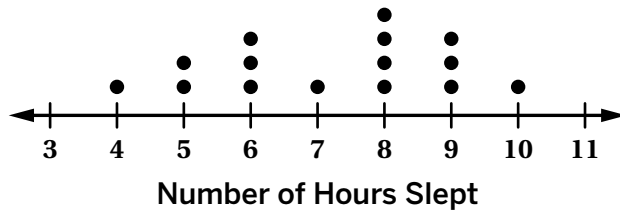
- 7.** Jada claims that Polygon A and Polygon B have the same perimeter. Is she correct? Explain your thinking.



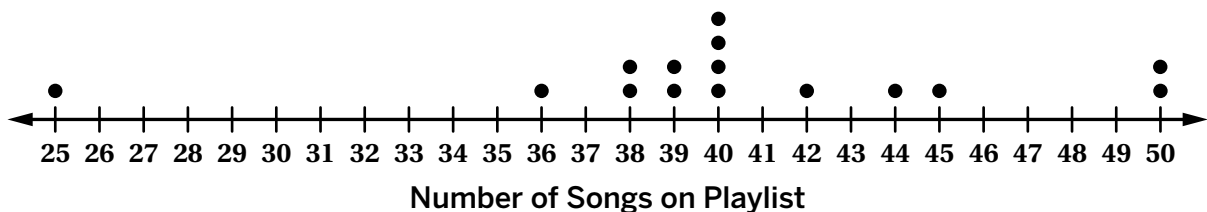
Additional Practice

ACC6.8.01

Problems 1–2: Here is a dot plot.

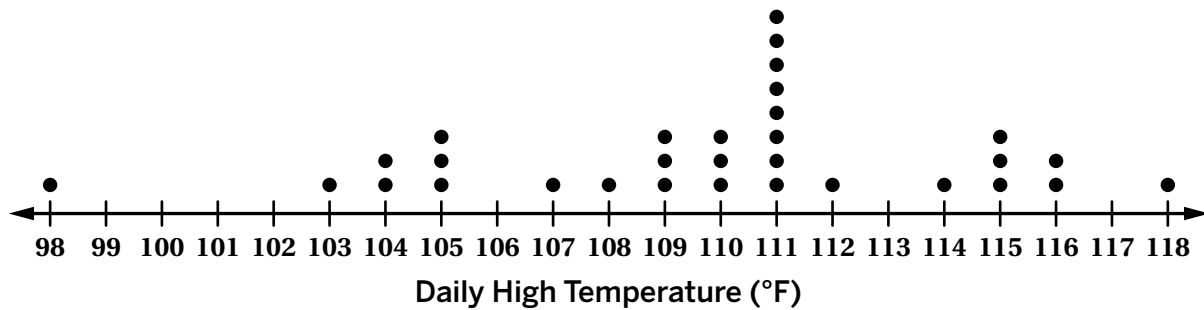


- Shawn surveyed several students to determine the number of hours they slept the previous night. How many students did Shawn survey?
- Shawn slept for 10 hours last night. He claims to have slept about the typical amount as all the other students surveyed. Is Shawn correct? Explain your thinking.
- A group of students was asked, “How many songs are on your favorite playlist?” The results are shown on this dot plot. Select *all* the statements that are true.



- ☐ A. 9 students were asked about the number of songs on their playlist.
- ☐ B. The most occurring number of songs on the playlist is 40.
- ☐ C. Typically, there are about 42 songs on a playlist.
- ☐ D. One-third of students had more than 40 songs on their playlist.
- ☐ E. More than half of the students had between 37 and 41 songs on their playlist.

Problems 4–5: This dot plot shows the daily high temperatures in Phoenix, Arizona, in July.

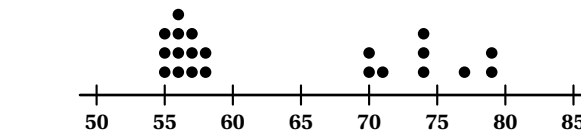
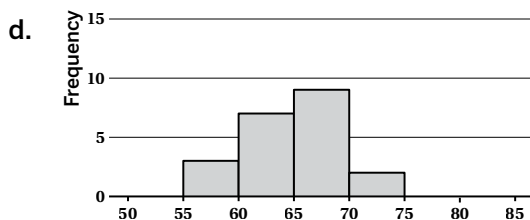
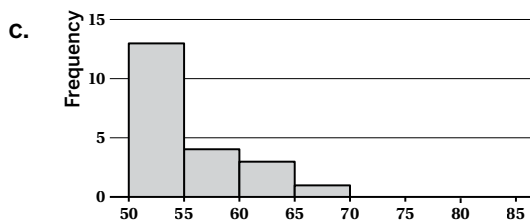
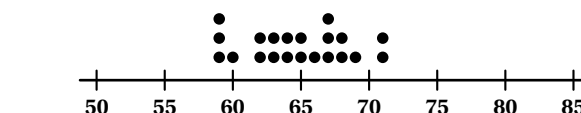
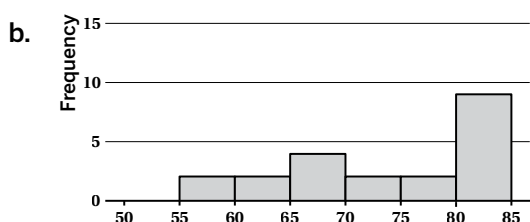
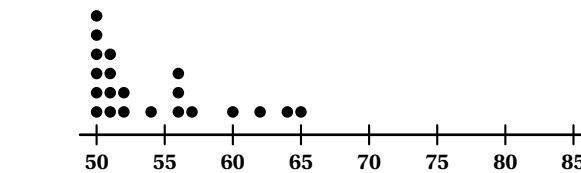
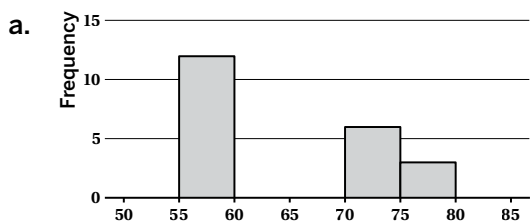


4. According to the dot plot:
- a What was the lowest daily high temperature?
 - b Which temperature occurred most often?
 - c How many days had a high temperature less than 110°F?
5. The high temperature on June 30 was 100°F. How does the temperature on June 30 compare to the temperatures in July? Select *all* that apply.
- ☐ A. The temperature on June 30 was much higher than the typical temperature in July.
 - ☐ B. The temperature on June 30 was within the range of temperatures in July.
 - ☐ C. The typical temperature in July was much higher than the temperature on June 30.
 - ☐ D. The minimum temperature in July was higher than the temperature on June 30.
 - ☐ E. There were thirteen days in July that had a higher temperature than June 30.
6. Select *all* the questions that are statistical questions.
- ☐ A. What is the shoe size of each student in a sixth grade class?
 - ☐ B. What is the eye color of a student in the cafeteria?
 - ☐ C. What modes of transportation do students at our school use to get to school each day?
 - ☐ D. How many soccer goals did a team score at each game during their season?
 - ☐ E. What are the overnight low temperatures for a city in Florida in the last month?
7. Clare said the question “What is the height of all the students in third grade?” is not a statistical question. Is Clare correct? Explain your thinking.

Additional Practice

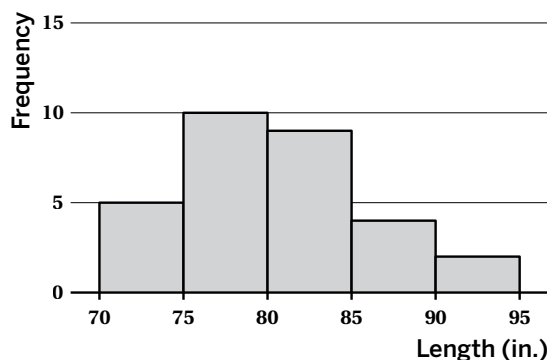
ACC6.8.03

1. Match each histogram with a dot plot that represents the same data set.



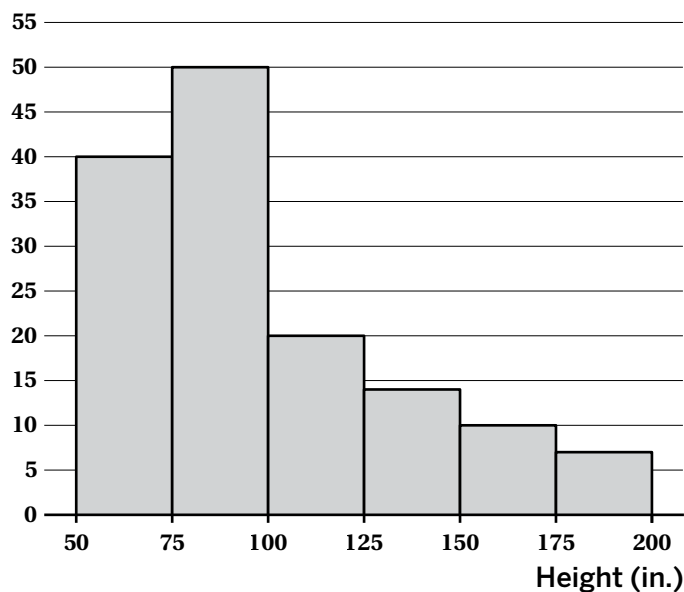
2. The histogram summarizes the lengths, in inches, of a group of West Atlantic bluefin tuna. Select *all* of the true statements.

- ☐ A. A typical tuna is between 75 and 85 inches long.
- ☐ B. The longest tuna was 70 inches long.
- ☐ C. 10 tuna were 95 inches long.
- ☐ D. A total of 30 tuna were measured.
- ☐ E. A total of 5 tuna were measured.
- ☐ F. Five tuna were less than 75 inches long.

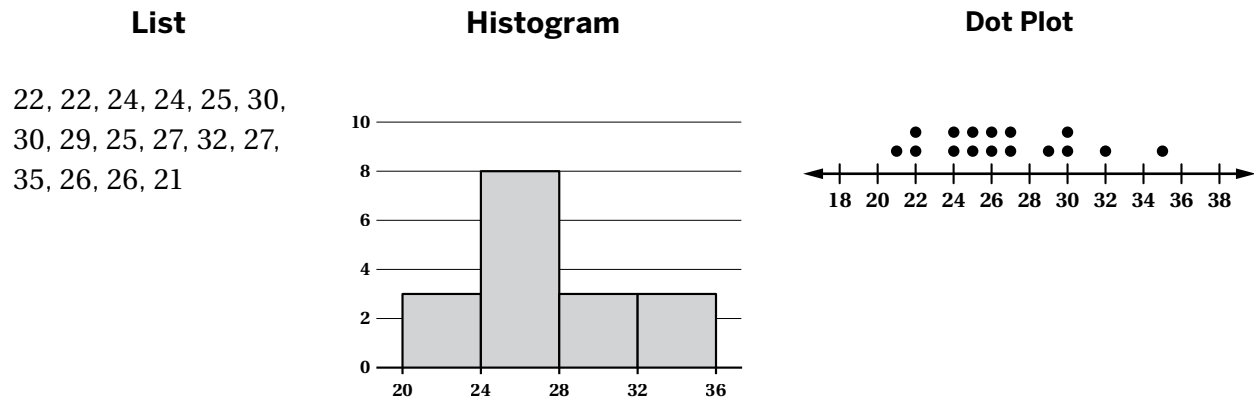


Problems 3–4: Here is a histogram showing the heights of peach trees, in inches, on an acre of land.

- 3.** How many peach trees are at least 75 inches tall but less than 125 inches tall?
- 4.** An average peach tree is between 144 and 180 inches tall. Bard says that there are 24 peach trees between these heights on this acre of land. Is Bard correct? Explain your thinking.



Problems 5–6: Here are three representations of the same data set.



- 5.** Circle the representation that doesn't match the other two.
- 6.** Revise your circled representation to make it match the other two representations.

Additional Practice

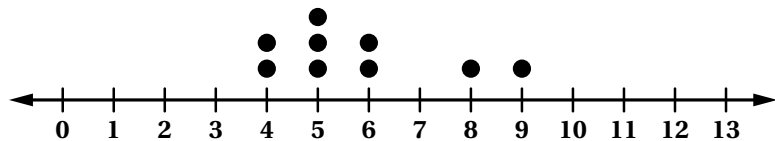
ACC6.8.04

1. This data set represents the daily high temperatures for Minneapolis, Minnesota, in degrees Celsius.

Degrees Celsius						
6	9	8	10	8	6	9

What was the average high temperature?

2. Without calculating, determine if the mean of the data set is 8. Explain your thinking.



3. The mean of four numbers is 12. Three of the numbers are 8, 12, and 12. What is the fourth number? Show or explain your thinking.

4. This data set represents the number of goals a soccer team scored at each game during their season.

Number of Goals						
3	5	8	2	4	3	1
0	1	1	5	3	2	4

What was the mean number of goals scored per game?

- 5.** An art teacher is rearranging four bags of popsicle sticks so that each bag contains an equal number of sticks. Currently Bag A has 45 sticks, Bag B has 25 sticks, Bag C has 16 sticks, and Bag D has 34 sticks. Select *all* the ways the art teacher could make each bag have the same number of popsicle sticks.
- ☐ **A.** Remove 4 popsicle sticks from Bag D and 15 popsicle sticks from Bag A. Redistribute the 19 sticks that were removed into Bags B and C so there are 30 sticks in each bag.
 - ☐ **B.** Remove 5 popsicle sticks from Bag D and place them in Bag B.
 - ☐ **C.** Remove 15 popsicle sticks from Bag A and place them in Bag B. Remove 5 popsicle sticks from Bag D and place them in Bag C.
 - ☐ **D.** Remove 15 popsicle sticks from Bag A and place them in Bag C. Remove 5 popsicle sticks from Bag D and place them in Bag B.
 - ☐ **E.** Remove all the popsicle sticks and make four equal piles, which will contain 30 sticks each. Then put each pile in one of the bags.
- 6.** Noah babysat 6 times. He earned \$24, \$25, \$31, \$32, and \$28 for the first 5 babysitting jobs. How much did Noah earn at the sixth babysitting job if the average amount he earned was \$27?
- 7.** In her math class, Priya's teacher gives 5 quizzes, each worth 10 points. After 4 of her quizzes, Priya has scores of 7, 10, 8, and 6. How many points does Priya have to score on the last quiz to have an average score of 8?
- 8.** While playing a card game, Shawn kept score for the first 5 hands, as shown in the table. Shawn claims that the mean score per hand is 15.

Score				
14	10	15	20	16

Is Shawn correct? If yes, explain how Shawn is correct. If not, explain how to calculate the correct mean.

Additional Practice

ACC6.8.05

Problems 1–4: This table shows the amount of time it takes 5 laptops to charge to full battery. The mean charge time is 30 minutes.

Charge Time (min)	30	31	29	29	31
Absolute Deviation					

- Complete the table by calculating the absolute deviation of each value from the mean.
- What is the sum of the absolute deviations?
 - 0
 - 3
 - 4
 - 5
- Calculate the mean absolute deviation (MAD) of this data set. Show or explain your thinking.
- Based on the calculated MAD, how are the data points spread around the mean?
 - The MAD is a smaller value, so the data points are *less* spread out around the mean.
 - The MAD is a smaller value, so the data points are *more* spread out around the mean.
 - The MAD is a larger value, so the data points are *less* spread out around the mean.
 - The MAD is a larger value, so the data points are *more* spread out around the mean.

Problems 5–7: Mia would like to know how long her classmates studied for an upcoming French exam. This table shows the amount of time 5 students studied for the exam. Their mean study time was 90 minutes.

Study Time (min)	80	92	111	140	27
Absolute Deviation					

5. Complete the table by calculating the absolute deviation of each value from the mean.

6. What is the mean absolute deviation (MAD) of this data set?

A. 28.7

B. 29.2

C. 30.4

D. 31.6

7. Describe how the calculated MAD measures the spread of the data around the mean study time.

Additional Practice

ACC6.8.06

1. Select *all* the true statements about the median.

- ☐ A. The median is the middle number.
- ☐ B. If there are 9 data points, calculate the average of the two in the middle.
- ☐ C. The data set needs to be in numerical order before determining the median.
- ☐ D. The median represents both a measure of center and a typical number.
- ☐ E. The mean and median are always the same.

2. This data set shows the number of tornadoes in Texas over several years.

89	184	58	186	106	258	47	84	121
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- a Order the data from *least* to *greatest*.
- b What is the median of this data?

3. Diego's score for each hole in mini golf is shown.

3	4	1	6	4	6	3	3	2
5	4	3	5	7	2	4	4	1

What was his median score?

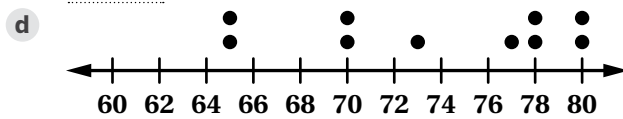
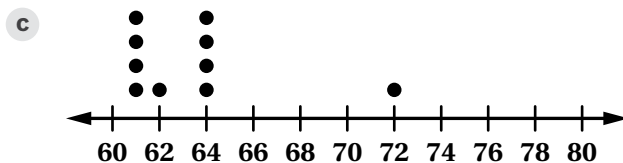
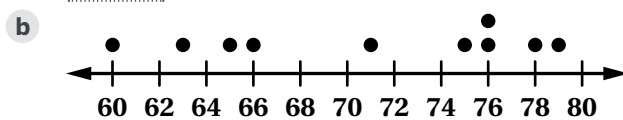
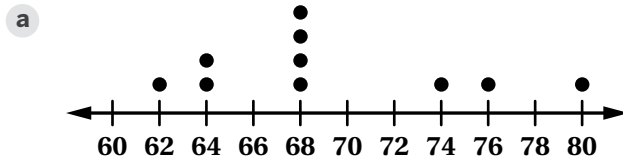
- A. 3
- B. 4
- C. 6
- D. 9

4. Shawn and Lin are reading a trilogy of books. The table lists the number of minutes each of them read on several days over the past few weeks.

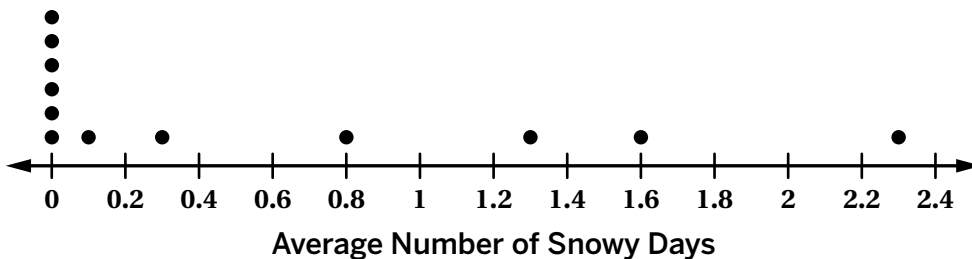
Shawn	30	30	25	15	40	35	20	20	30	25	35	40
Lin	15	15	25	20	15	40	30	40	40	30	25	20

- a Determine the median of each data set.
- b Who typically reads more? Explain your thinking.

5. Calculate the median for each dot plot.



6. This dot plot shows the average number of snowy days each month for Portland, Oregon. Determine the median for the data set.



7. Elena is researching backpacks. She sorted the price of several backpacks and noticed that 6 backpacks were more expensive than the one she purchased, and 8 backpacks were less expensive than the one she purchased. Does this mean that the price of Elena's backpack is the median? Explain your thinking.

Additional Practice

ACC6.8.07

1. This data set represents the scores of several students in a class on a 10-point quiz.

5 5 6 7 7 7 8 9 9 9 9 10 10 10 10

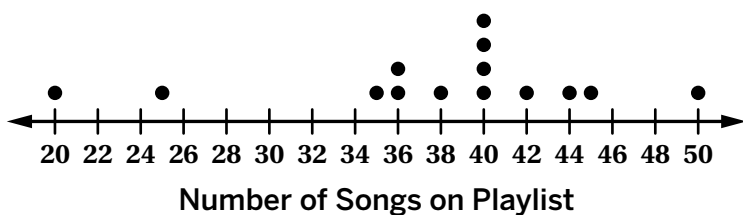
- a Determine the value of the median (Q2).
- b Determine the value of Quartile 1 (Q1).
- c Determine the value of Quartile 3 (Q3).

2. This data set represents the number of goals a soccer team scored at each game during their season.

3 5 8 2 4 3 1 0 1 1 5 3 2 4

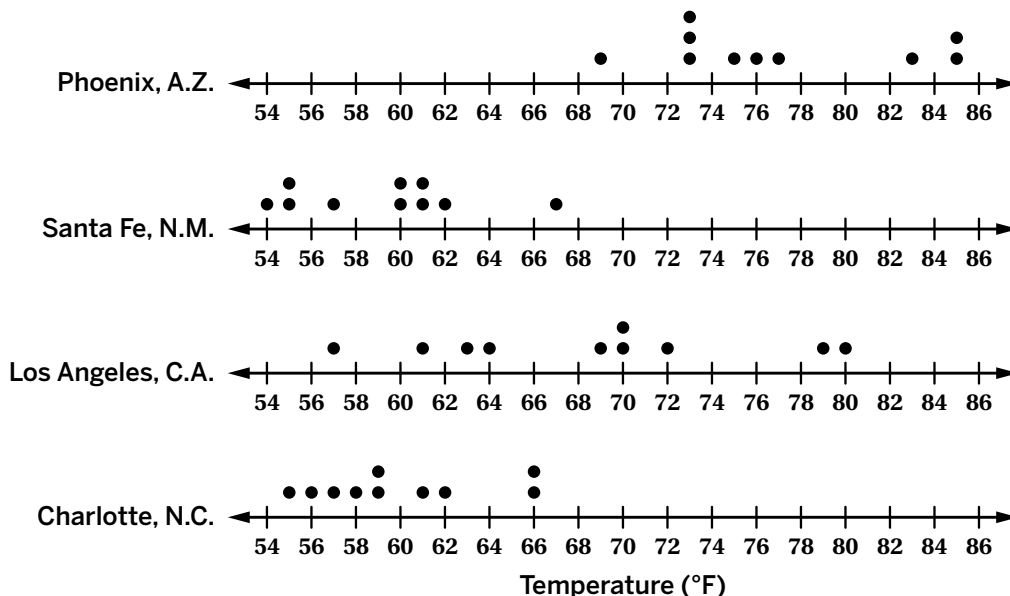
- a Determine the value of the median (Q2).
- b Determine the value of Quartile 1 (Q1).
- c Determine the value of Quartile 3 (Q3).

3. A group of students was asked, "How many songs are on your favorite playlist?" The results are shown on this dot plot.



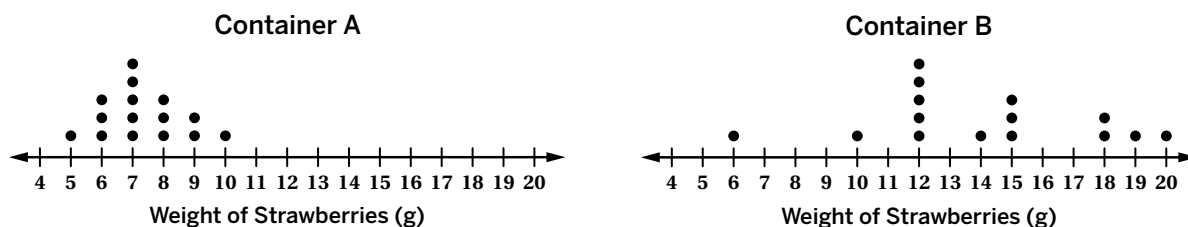
- a Determine the value of the median (Q2).
- b Determine the value of Quartile 1 (Q1).
- c Determine the value of Quartile 3 (Q3).

4. These dot plots represent the daily high temperatures in degrees Fahrenheit for four different cities in March. Match each dot plot with the correct quartiles by writing the city's name next to each set of statistics.



- a Median: 59, Q1: 57 b Median: 60, Q3: 61
- c Median: 69.5, Q3: 72 d Median: 75.5, Q1: 73

Problems 5–6: Tyler weighs strawberries from two different containers and records the weights on the dot plots.



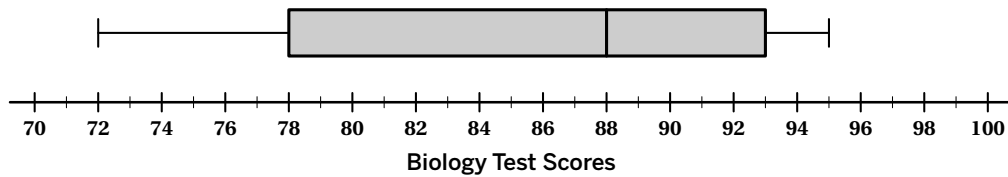
5. Explain how you can determine that Container B has a greater median (Q2) than Container A, without calculating the values.
6. Determine the values for Q1, Q2 (median), and Q3 for each container.

	Q1	Q2 (median)	Q3
Container A			
Container B			

Additional Practice

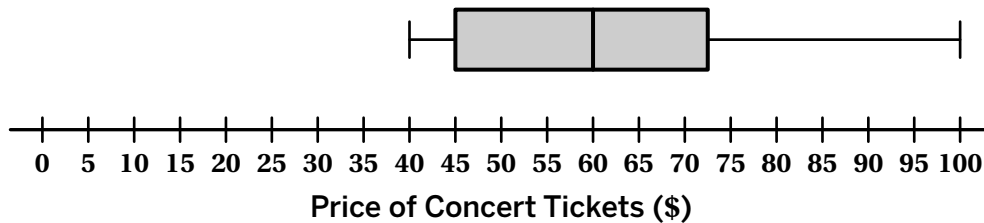
ACC6.8.08

1. Each student in a class recorded their score on a recent biology test.



- a What is the greatest score in this class?
- b What is the median of this data?
- c What is the range of this data?

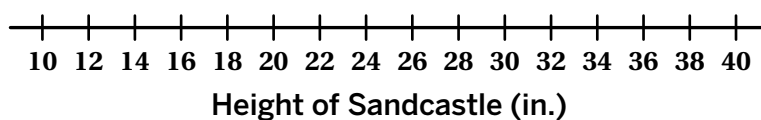
2. A venue recorded ticket prices for 250 tickets sold for a concert.



- a What percent of tickets sold were between \$60 and \$100?
- b What percent of tickets sold were under \$60?
- c What percent of tickets sold were over \$45?
- d How many tickets sold were between \$45 and \$100?

3. The data shows the height, in inches, of several sandcastles built on the beach. Create a box plot to represent this data set.

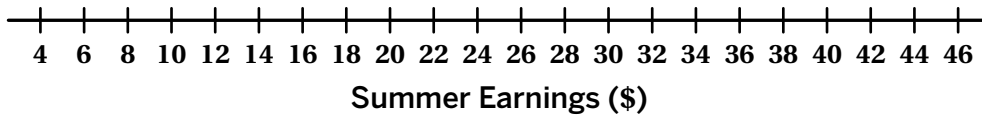
18	34	30
28	15	30
36	24	22



Problems 4–6: The data shows the amount of money Diego earned for doing odd jobs over the course of the summer.

Minimum: 10 **Q1:** 17 **Median:** 25 **Q3:** 30 **Maximum:** 45

- 4.** Create a box plot above the number line to represent the data.



- 5.** Which of the following statements are true about the box plot? Select *all* that apply.
- ☐ **A.** The IQR is 15.
 - ☐ **B.** About half of Diego's jobs paid between \$17 and \$30.
 - ☐ **C.** Diego's highest-paying job paid \$45.
 - ☐ **D.** About 50% of the jobs paid Diego more than \$23.
 - ☐ **E.** About 25% of the jobs paid Diego less than \$30.
- 6.** Diego says that 25% of the jobs paid between \$25 and \$40. Is Diego correct? Explain your thinking.

Additional Practice

ACC6.8.10

1. Identify each statement as *true* or *false*. Explain your thinking.

- a When information is being gathered about a group, the entire group is called the sample.
- b Different random samples of the same size and from the same population will be the same.

2. The student council at a local middle school surveyed a random sample of 100 students to see how they felt about the lunch offerings. Identify the population and sample in this scenario.

- A. The population is all of the seventh graders at the middle school. The sample is the students surveyed.
- B. The population is all of the students surveyed. The sample is all of the students at the school.
- C. The population is all of the students at the school. The sample is the students surveyed.
- D. The population is all of the students at the school. The sample is all of the seventh graders.

Problems 3–4: A bakery makes hundreds of muffins each day, including bran, blueberry, and cinnamon muffins. They decide to survey a random sample of their customers to determine which muffin tastes best to customers.

3. Which statement *best* describes the population?

- A. The population is all of the bakers at the bakery.
- B. The population is all of the customers of the bakery.
- C. The population is all of the bran, blueberry, and cinnamon muffins.
- D. The population is one kind of muffin made at the bakery.

4. Which statement *best* describes the sample?

- A. The sample is all of the muffins made at the bakery.
- B. The sample is one bran muffin, one blueberry muffin, and one cinnamon muffin.
- C. The sample is all of the customers that come into the bakery.
- D. The sample is a randomly selected group of customers.

Problems 5–7: Jamya wonders: *What time do middle school students at my school go to sleep?*

5. What is the population for Jamya's question?

6. Select *all* possible samples for Jamya's question.

- ☐ **A.** The 25 students in a 6th grade math class
- ☐ **B.** The first 20 people to arrive at the middle school on a particular day
- ☐ **C.** 2 randomly selected students from each math class at the middle school
- ☐ **D.** All the students sitting at a randomly selected table during an 8th grade lunch period
- ☐ **E.** The 20 students in a high school science class

7. Select two samples and list a different possible population each sample could belong to.

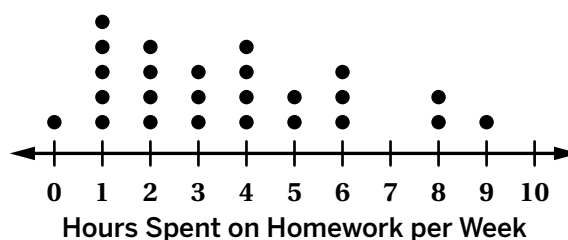
Sample	Possible Population
The 25 students in a 6th grade math class	
The first 20 people to arrive at the middle school on a particular day	
2 randomly selected students from each math class at the school	
All the students sitting at a randomly selected table during an 8th grade lunch period	
The 20 students in a high school science class	

Additional Practice

ACC6.8.11

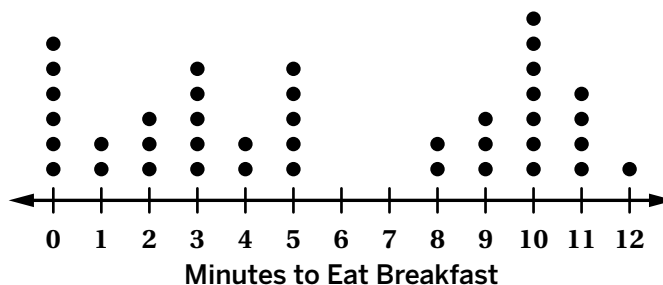
- Lin's school had a fundraiser where about 60% of students paid \$1 to wear a hat all day at school. Lin selects a sample of 45 students. Her sample showed that 80% of students paid \$1. Is this a representative sample? Explain your thinking.

- Tyler was asked how many hours students at his school spend on homework each week. The sample shown consists of 25 students and is representative of the population.



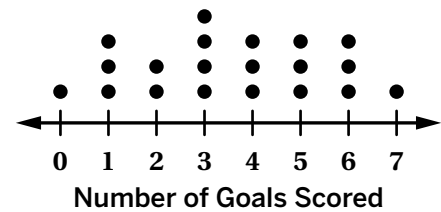
Because this sample is representative of the population, what might a dot plot for the entire population look like? Select *all* that apply.

- ☐ A. The center will be around 4.5.
 - ☐ B. Very few data values will be below 8.
 - ☐ C. The population will have more data on the left than on the right.
 - ☐ D. The range will start at 0.
- Shawn was asked how long it takes students at school to eat breakfast. The sample shown consists of 40 students and is representative of the population. Complete these sentences to describe what a dot plot for the entire population would look like.



- There will be few dots near _____ and _____.
- The center will be around _____. (Round to the nearest tenth.)

Problems 4–5: Priya surveyed soccer players at her school to see how many goals they scored during their most recent soccer season. The results of the survey are shown on the dot plot.



4. Calculate the mean and MAD for this data set.

a Mean

b MAD

5. Priya thinks that her survey took too long. She decides that next time, she will survey a sample of the soccer players. She plans to ask the soccer team captains from each grade level. Do you think this method is likely to produce a representative sample? Explain your thinking.

Problems 6–8: There are 24 students in Tyler's math class. The average test score on the last test was 91.

6. Mai scored 100 on the test, and Noah scored 98. Are their scores representative of the 24 students? Show or explain your thinking.

7. Diego scored 81 on the test, Clare scored 94 on the test, and Tyler scored 98 on the test. Are their scores representative of the 24 students? Show or explain your thinking.

8. Andre scored 90 on the test, Elena scored 88 on the test, Han scored 90 on the test, and Jada scored 92 on the test. Are their scores representative of the 24 students? Show or explain your thinking.

Additional Practice

ACC6.8.12

- Diego surveyed a random sample of 20 students and asked them whether they prefer math or science. 12 students said they preferred math. Clare did not think Diego's sample was representative of the population, so she surveyed a random sample of 80 students, and 42 said they preferred math.
 - Based on Diego's sample, estimate what fraction of the students preferred math.
 - Based on Clare's sample, estimate what fraction of the students preferred math.
 - Whose estimate is more likely to be accurate? Explain your thinking.

Problems 2–4: Andre and Elena surveyed a random sample of 24 students at each of their schools, asking each student how many people are in their household.

Andre's Sample

6	2	6	8	2	3
4	3	3	7	3	6
6	4	5	4	5	4
5	8	5	4	4	4

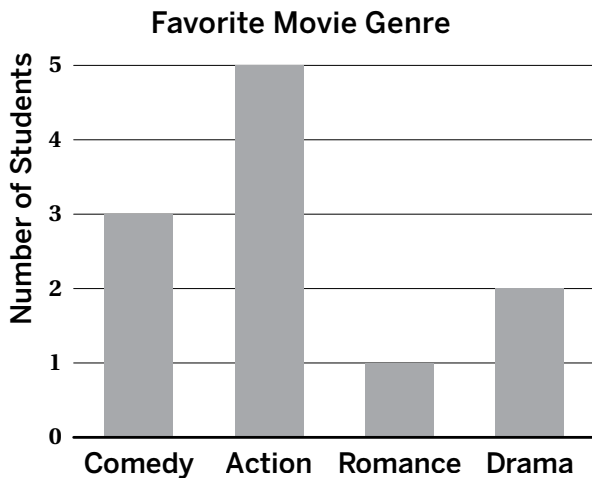
Elena's Sample

4	4	3	6	4	6
4	4	5	6	2	4
3	6	4	3	5	2
5	2	8	4	5	4

- For each sample, what fraction of the students have 4 people in their household?
 - Andre's sample:
 - Elena's sample:
- There are 1,050 students at Andre's school. Estimate the number of students at Andre's school who have the following number of people in their household:
 - 3 people in their household
 - 4 people in their household
- There are 975 students at Elena's school. Estimate the number of students at Elena's school who have the following number of people in their household:
 - 6 people in their household
 - 4 people in their household

Problems 5–6: There are 978 students at Han’s school. Han surveyed a random sample of students about their favorite movie genre. The bar graph shows the results.

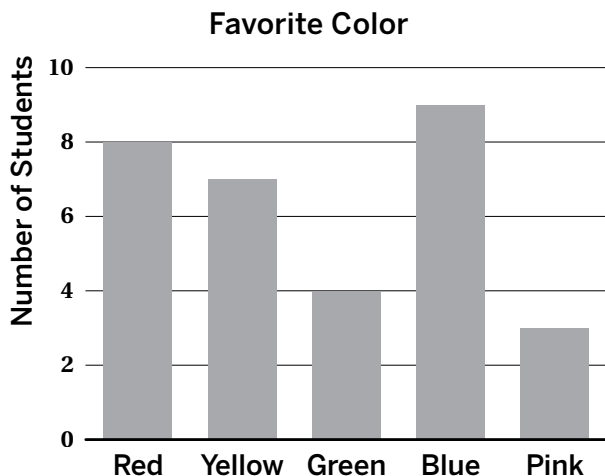
- Estimate the total number of students in the school who would choose action as their favorite movie genre.



- Estimate the total number of students in the school who would choose romance or drama as their favorite movie genre.

Problems 7–8: There are 868 students at Shawn’s school. Shawn surveyed a random sample of students about their favorite color. The bar graph shows the results.

- Shawn estimates that 196 students would choose yellow as their favorite color. Do you think Shawn’s estimate is likely to be accurate? Explain your thinking.



- Shawn estimates that 504 students would choose red or blue as their favorite color. Do you think Shawn’s estimate is likely to be accurate? Explain your thinking.

Additional Practice

ACC6.8.13

Problems 1–3: Clare loves to read. This data shows the number of pages Clare read each week during her winter and spring semesters.

Winter				
260	200	250	240	290
200	190	190	250	300
MAD: 33.6				
Spring				
200	260	250	240	280
190	210	200	250	260
MAD: 27.2				

1. Determine the mean number of pages that Clare read during her winter and spring semesters.
2. Calculate how many MADs apart the means are. Use the larger MAD in your calculation.
3. Based on this data, did Clare read more pages during winter than during spring? Explain your thinking.

Problems 4–5: Han compared the time that students and teachers spent driving to school each day over one week. He took a random sample of 20 students and 20 teachers. The results are shown in the table.

	Mean (min)	MAD (min)
Students	22.7	8.3
Teachers	34.5	7.4

- 4.** How many MADs apart are the means? Use the larger MAD in your calculation.
- A. 1.42 MADs
 - B. 1.59 MADs
 - C. 6.89 MADs
 - D. 7.73 MADs
- 5.** Is there a big difference between the students' data and the teachers' data? Circle one.
- Yes No Maybe

Explain your thinking.

Additional Practice

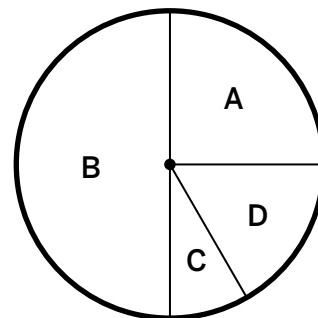
ACC6.8.15

1. Order these events from *least likely* to *most likely*.

Landing on A, Landing on B, Landing on C, Landing on D

Least Likely

Most Likely



Problems 2–4: List the sample space for each experiment.

2. Selecting a random month of the year
3. Selecting a random letter from the word CHAIR
4. Flipping a coin:

Problems 5–7: Determine the probability of each event.

5. A month starting with J is selected at random.
6. Selecting a vowel at random from the word CHAIR
7. A fair coin lands on tails

Problems 8–13: Determine the probability of each event.

8. Selecting a white marble from a bag containing 5 red marbles and 6 black marbles.

9. Selecting a vowel from the word LOOM.

10. Rolling a 6 on a 10-sided die.

11. A spinner has 6 equal-sized sections labeled 1 through 6. You spin the spinner and it lands on an even number.

12. Selecting a black marble from a bag containing only black marbles.

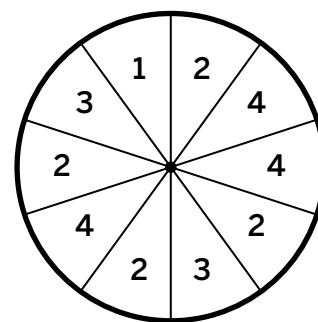
13. Selecting a consonant from the word GRADE.

14. A spinner has 10 equal-sized sections. Order these events from *least likely* to *most likely*.

Landing on 1, Landing on 2, Landing on 3, Landing on 4

Least Likely

Most Likely



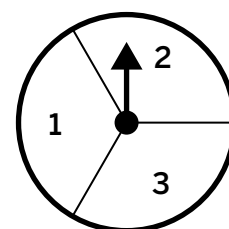
15. A letter will randomly be selected from the word ALGEBRA. Determine the probability that the letter A will be chosen.

Additional Practice

ACC6.8.16

Problems 1–2: A day is chosen at random from the days of the week.

1. What is the probability of getting a day that starts with the letter S? Consider listing the sample space to show your thinking.
2. If you repeat this experiment 700 times, how many times do you expect to get a day that starts with S?
3. A bag has 10 blocks in it. A student picks a block out of the bag 50 times and gets a blue block 16 times. Based on these results, how many blocks do you expect to be blue? Explain your thinking.
4. Braylen spun this spinner 60 times. How many times would you expect the spinner to land on 1?



Problems 5–7: There are 365 days in a year. One day is chosen at random.

5. What is the probability that it is November 12th?

6. What is the probability that it is one of the days in November?

Note: There are 30 days in November.

7. If you repeat this experiment 100 times, how many times do you expect to choose one of the days in November?

8. The table shows the genres of books on a shelf. A student will randomly choose one book from the shelf.

Based on the information in the table, which statement is true:

- A.** The book is least likely to be non-fiction.
- B.** The book is equally likely to be poetry as it is to be a mystery.
- C.** The book is most likely to be a graphic novel.
- D.** The book is twice as likely to be non-fiction as it is to be poetry.

Genre	Number of Books
Fantasy	1
Non-fiction	6
Mystery	5
Poetry	2
Graphic novel	10

Additional Practice

ACC6.8.17

1. Aniyah rolls a number cube 100 times. She rolls a three 29 times. She decides that it is not a standard number cube.

a Do you agree with Aniyah? Explain your thinking.

b What could Aniyah do to be more sure of her results?

2. Demari made a spinner and spun it 300 times. It landed on a yellow section 225 times. In Demari's experiment, what was the relative frequency of spinning yellow? Select *all* that apply.

- | | | |
|---|---|---|
| ☐ A. $\frac{1}{3}$ | ☐ B. 0.25 | ☐ C. 33% |
| ☐ D. 0.75 | ☐ E. $\frac{3}{4}$ | ☐ F. 25% |
| ☐ G. 75% | ☐ H. 0.33 | ☐ I. $\frac{1}{4}$ |

3. Kyle has a bag containing 2 red blocks and 3 blue blocks. She draws a block out of the bag, then replaces it. Kyle plans to repeat the experiment 150 times. Which statement best predicts how many times Kyle will draw a blue block?

- A. She will draw a blue block exactly 90 times.
- B. She will draw a blue block around 90 times but probably not exactly 90 times.
- C. She will draw a blue block exactly 100 times.
- D. She will draw a blue block about 50 times.

4. Adrian has a coin he suspects is not fair. Describe an experiment Adrian could use to determine whether the coin is fair.

5. Middle school students played a game where they spun a spinner to win a copy of one of four books. The results are shown in the table. What is the relative frequency of the Dragon's Land series, rounded to the nearest percent?

- A. 30%
B. 34%
C. 23%
D. 36%

Book Title	Frequency
Alex's Mysteries	28
The Soldier's Apprentice	36
The Land of Games	16
Dragon's Land	42

Problems 6–8: Amal flips a coin 400 times. It lands on heads 212 times.

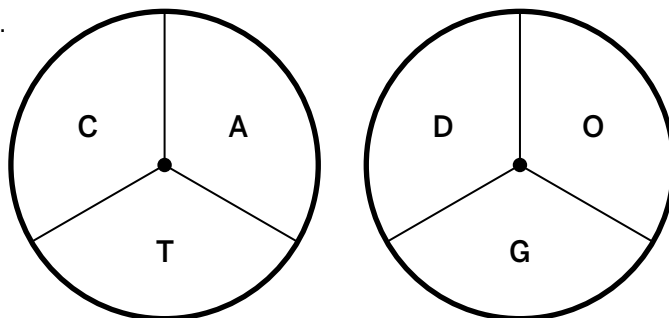
6. What is the relative frequency of heads in Amal's experiment?
7. Based on these results, is it likely that Amal's coin is fair? Explain your thinking.
8. If Amal flipped the coin another 70 times, about how many times would you expect it to land on heads?

Additional Practice

ACC6.8.18

1. Shawn spins the two fair spinners shown.

- a List *all* of the possible outcomes.
- b How many different possible outcomes are in the sample space?



2. Priya tosses a dime, rolls a standard number cube, and then tosses a penny. Use the table to list all of the possible outcomes.

Note: Use H for heads and T for tails.

3. Here are two tables. For each event, use any method to determine the sample space. Then determine the number of outcomes.

- a Andre selects one type of milk and one type of cereal to make breakfast.

Milk	Cereal
Dairy	Granola
Soy	Oatmeal
	Rice squares

- b Andre selects one type of base, one type of protein, and one type of vegetable to make a lunch bowl.

Base	Protein	Vegetable
Rice	Chicken	Tomatoes
Lettuce	Beans	Corn

4. Clare rolls a standard number cube and spins a spinner with the letters G, O, E, and S on it. She claims there are 10 possible outcomes. Is Clare correct? Explain your thinking.

Problems 5–8: A breakfast diner makes omelettes with one type of egg, one protein, one type of cheese, and one vegetable. Customers can choose from the options shown in the table.

Eggs	Protein	Cheese	Vegetables
Whole eggs	Bacon	American	Onions
Egg whites	Sausage	Mozzarella	Peppers
	Tofu	None	Mushrooms
			Tomatoes

5. How many different omelettes are possible, assuming customers must choose one type of egg, one protein, one type of cheese, and one vegetable?
6. If a customer chooses a protein and a vegetable at random, is getting tofu or mushrooms more likely? Explain your thinking.
7. Tyler wants an omelette that has egg whites and tomatoes. He does not have a preference for a protein or cheese. How many different omelettes could Tyler choose? Show or explain your thinking.
8. Suppose an omelette is made by randomly choosing each of the options. What is the probability that the omelette will be what Tyler wants?

Additional Practice

ACC6.8.19

1. Mai surveyed students in her class about their homework. 8 students had homework last night, and 4 students did not have homework last night. Mai used this data to design a spinner with 6 equal-sized sections that predicts whether a randomly selected student had homework last night. How many sections of Mai's spinner represent students who did not have homework?

A. 2 B. 4 C. 6 D. 8

2. A restaurant owner found that 80% of her customers return to the restaurant within one month. She wants to simulate the habits of her customers to help her predict the probability that three of the next four customers to visit her restaurant will return within one month. Which simulation can be used to find the indicated probability?

A. A spinner divided into 2 equal-sized sections, spun 4 times

B. A spinner divided into 2 equal-sized sections, spun 3 times

C. A spinner divided into 5 equal-sized sections, spun 4 times

D. A spinner divided into 5 equal-sized sections, spun 3 times

3. 12 of the 20 students in Lin's class are girls. Each month, Lin's teacher randomly selects a student to be an assistant. Which simulation can be used to find the probability that a girl will be chosen for August, September, October, and December?

A. A bag of 15 marbles, with 9 marbles representing the girls, has a marble drawn 4 times with replacement.

B. A number cube, with odd numbers representing girls, is rolled 12 times.

C. A coin, with heads representing girls, is tossed 4 times.

D. A spinner, divided into 4 equal-sized sections with three of the sections representing girls, is spun 12 times.

4. 15 of the last 20 winners of the school spelling bee have been seventh graders. Noah placed 3 yellow marbles and 1 red marble into a bag to create a simulation to predict whether future spelling bee winners will be seventh graders. According to Noah's model, what is the probability that the next 4 winners will be seventh graders?

A. $\frac{3}{4}$ B. $\frac{9}{16}$ C. $\frac{27}{64}$ D. $\frac{81}{256}$

5. The students at a local middle school are divided into four homerooms. A new student will be randomly assigned to one of the homerooms. Which methods can be used to simulate this scenario? Select *all* that apply.
- ☐ A. Spin a spinner with 8 equal-sized sections with 2 sections assigned to each homeroom.
 - ☐ B. Roll a number cube with 1 representing Homeroom 1, 2 representing Homeroom 2, 3 representing Homeroom 3, and 4, 5, and 6 representing Homeroom 4.
 - ☐ C. Draw a marble from a bag containing 3 white marbles, 3 blue marbles, 3 yellow marbles, and 3 green marbles, with each color representing a different homeroom.
 - ☐ D. Toss a coin with heads representing Homerooms 1–3 and tails representing Homeroom 4.

6. Priya's family cat just had two kittens. Priya wants to know the probability of a cat having two kittens in which one is male and one is female. To estimate the probability, Priya designed and conducted a simulation using a coin toss. She let heads represent a female kitten and tails represent a male kitten. Her results are shown in the table.

Trial (toss)	1	2	3	4	5	6	7	8	9	10
Results	HH	HT	TT	HT	HT	TT	HH	HH	TT	HT

- a How many trials did Priya conduct?
 - b Based on this simulation, estimate the probability that a cat having two kittens will have one male and one female.
 - c Based on this simulation, estimate the probability that a cat having two kittens will have two females.
 - d Based on this simulation, estimate the probability that a cat having two kittens will have two males.
7. A class quiz has 6 multiple choice questions, and each question has 4 possible answers. Shawn and Tyler want to simulate the score of a student who guessed randomly on each question. Shawn claims it would be best to simulate this scenario with a spinner divided into 6 equal-sized sections. Tyler claims it would be best to simulate this scenario with a spinner divided into 4 equal-sized sections. Who is correct? Explain your thinking.